Process Characterization Report

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCR-007

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

Table of contents

Approvals 5
Revision history 6
Abbreviations 6

Executive summary **8**

1 Introduction **10**
1.1 Product and unit operation 10
1.2 Objectives 11
1.3 Regulatory and scientific basis 11
1.4 Changes in this revision 12

2 Prior knowledge and quality risk basis **15**
2.1 Platform and prior-product knowledge 15
2.2 Quality attributes in scope 16
2.3 Risk-based prioritization and parameter selection 17

3 Materials and methods **18**
3.1 Scale-down model and its qualification 18
3.2 Operation 20
3.3 Analytical methods 21
3.4 Sampling plan 23
3.5 Statistical methods 24
3.5.1 Decisions for which the plan sets no threshold 27

4 Study design **29**
4.1 Factors, ranges and the knowledge space 29
4.2 Screening design 29
4.3 Response-surface design 30
4.4 Univariate assessment 30

5 Results **31**
5.1 Centre-point performance and reproducibility 31
5.2 Factor effects in the screening design 31
5.3 Response-surface models 32
5.3.1 Aggregate 34
5.3.2 Host cell protein 36
5.3.3 Influential runs 38
5.3.4 Models within two AICc units of the selected model 39
5.3.5 Fragments 39
5.3.6 Step yield 40

5.3.7 Response surfaces . . . 40
5.4 Residual DNA and leached Protein A . . . 41
5.5 Univariate assessment of elution flow rate . . . 41
5.6 Mechanistic interpretation . . . 43

6 Design space **45**

7 Proven acceptable ranges **46**
7.1 Acceptance basis . . . 46
7.2 Outputs against their criteria . . . 48
7.3 Proven acceptable ranges per parameter . . . 50
7.4 Verification of the platform worst cases . . . 52

8 Process capability and robustness **55**

9 Parameter classification **57**

10 Contribution to the control strategy **59**

11 Discussion **60**

12 Conclusions **62**

13 Study events and discrepancies from the plan **64**
13.1 Study events . . . 64
13.1.1 DEV-007-01 Expired buffer lot in a screening run . . . 64
13.1.2 DEV-007-02 Temperature excursion during a response-surface run 65
13.2 Discrepancies from the plan . . . 65
13.2.1 PD-007-01 Model selection . . . 71
13.2.2 PD-007-02 Conclusions on a model that does not meet the residual criterion . . . 72
13.2.3 PD-007-03 Flow-rate study without the set-point run . . . 72
13.2.4 PD-007-04 Residual DNA offset recorded, not investigated . . . 72
13.2.5 PD-007-05 Load aggregate and host cell protein not recorded . . . 72
13.2.6 PD-007-06 Residual DNA and leached Protein A not measured on the screening runs . . . 73
13.2.7 PD-007-07 Flow-through, wash and strip fractions and the mass balance not recorded . . . 73
13.2.8 PD-007-08 Packing tests not recorded for each campaign . . . 73
13.2.9 PD-007-09 Fractionated runs not in the plan . . . 73
13.2.10 PD-007-10 Approaches where the plan sets no threshold, and step-level capability . . . 73
13.2.11 PD-007-11 Feed lots not recorded . . . 74
13.2.12 PD-007-12 Load/wash buffer named Tris in Version 1.0 . . . 74

Appendix A. Screening design matrix and responses **75**

Appendix B. Response-surface design matrix and responses **76**

Appendix C. Full effect and coefficient tables **78**

Appendix D. Analytical methods summary **80**

References **81**

Field	Value
Document ID	PCR-007
Document class	Process Characterization Report
Title	Process Characterization Report: Cation Exchange Chromatography (Step 7)
Product	A-Mab
Version	2.0
Effective date	2026-09-30
Supersedes	PCR-007 Version 1.0 (2026-07-24)
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
AICc	Akaike information criterion with the small-sample correction
AMD	analytical method decision
AMV	analytical method validation
AU	absorbance unit
BLA	Biologics License Application
CEX	cation exchange chromatography
CI	confidence interval
Cpk	process capability index
CQA	critical quality attribute
df	degrees of freedom
DNA	deoxyribonucleic acid
DoE	design of experiments
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
IgG1	immunoglobulin G, subclass 1
LMW	low molecular weight species
LOQ	limit of quantitation
Mfg	manufacturing scale
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
OD280	optical density at 280 nm
PAR	proven acceptable range
PPQ	process performance qualification

Abbreviation	Meaning
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
RSD	relative standard deviation
SD	standard deviation
SDM	scale-down model
SE	standard error
SEC	size exclusion chromatography
SOP	standard operating procedure
UF/DF	ultrafiltration/diafiltration
VI	low-pH viral inactivation
WC-CPP	well-controlled critical process parameter

Executive summary

Cation exchange chromatography is Step 7 of the A-Mab drug substance process and its principal aggregate polishing step. The antibody is bound to the resin and eluted by a step change in pH, and the step clears host cell protein, residual DNA, leached Protein A and part of the fragments. It sets no quality attribute of its own. This report presents the Stage 1 characterization of the step under the plan PCP-007 (U.S. Food and Drug Administration 2011). The characterization studies ran from 2026-02-04 to 2026-03-27 on a scale-down model of the commercial column, for a process that runs at a production bioreactor scale of 15,000 L at the commercial site in Grafton, WI. During the re-analysis, two studies of the low-pH viral inactivation report ran this step on the same scale-down model, at its set-point and on material from the low-pH step, to test how well it clears aggregate and fragments. They are S-CEX-VER, from 2026-09-24 to 2026-09-25, and S-VI-WC, from 2026-09-26 to 2026-09-29.

Protein load, load/wash conductivity, elution buffer pH and elution stop collect were studied in a two-level full factorial screening design of 19 runs and in a face-centred central composite design of 28 runs. Elution flow rate was assessed one factor at a time. The scale-down model is qualified for aggregate and host cell protein by its offsets from manufacturing scale, and every judgement of those two outputs is stated at manufacturing scale. Each output is judged against an intermediate acceptance criterion. The a priori criterion is the value derived before the studies, from the drug substance criterion and the platform worst cases of the later steps. The criterion in force is that value after this campaign’s data on the later steps tightened or widened it. For host cell protein the qualification holds at the criterion in force of 418 ng/mg and would not hold at the a priori criterion of 400 ng/mg.

Aggregate and host cell protein met their intermediate acceptance criteria at every combination of NOR low, set-point and NOR high. The single-run limit is the upper end of the 95 % prediction interval of a single run, at manufacturing scale. Its highest value over the NOR was 1.40 % for aggregate against 4.90 %, and 289 ng/mg for host cell protein against 418 ng/mg. The aggregate criterion is the drug substance criterion of 5.00 % less the largest rise expected across UF/DF, 0.104 percentage points. Fragments have no adequate model and were judged on their measured results, which lay between 0.47 % and 0.76 % against 2.00 %. On the single-run limit and at the criteria in force, the design space covers the whole characterized region. At the a priori host cell protein criterion it would cover 99.85 % of it. The proven acceptable range of every parameter for aggregate and host cell protein is its range studied.

The models of aggregate and host cell protein were selected on the log scale and meet the first three adequacy criteria of PCMP-001 §7.2. The host cell protein model does not meet the fourth, because its residuals drift with the order of execution. That drift

amounts to a factor of 1.13 over the study. The highest single-run limit over the NOR stays below the criterion with the whole drift added to it. The highest limit over the whole region studied would not, and nor would the limit over the NOR with the offset at the unfavourable end of its interval, on which the qualification rests (Section 5.3.2). Of the 5 parameters, 4 were classified as well-controlled critical process parameters. Elution flow rate was classified as a general process parameter, by a judgement of the practical relevance of its effect on residual DNA (Section 9).

Within its NOR the step cleared aggregate less than the platform worst case on which the criteria upstream rely. At manufacturing scale the lowest clearance factor predicted was 1.58 against 2.00. The one-sided 95 % prediction bound of a single batch, 1.40, replaced the platform value and tightened the aggregate criteria of the low-pH pool and the bioreactor harvest. The platform worst case for host cell protein was confirmed with the offset at its estimate. It would not be confirmed with the offset at the unfavourable end of its interval, or under 3 of the 5 other models within 2 AICc units of the selected model. The platform worst case for fragments was confirmed by the rule of PCP-007 §7.6, which confirms a platform worst case when the lowest clearance factor predicted within the NOR is not below it. The fragment model behind that prediction does not meet PCMP-001 §7.2. Leached Protein A is not a response of the designs.

The capability indices of the drug substance attributes that the step clears all exceed the target of 1.33, and the lowest is 6.14 for host cell protein. No simulated batch of 2,000 fell outside its criterion. The capability of the step itself, which the plan asked for, was not estimated.

Version 1.0 recorded 2 events of execution, which this report keeps as study events under their Version 1.0 DEV numbers. Screening run 6 used a load/wash buffer from an expired lot (DEV-007-01), and the column inlet temperature rose above its set-point during response-surface run 14 (DEV-007-02). Both runs were retained in the analysis. The register records 12 discrepancies from the plan. Four concern the analysis and the scale-down model. They are the model selection, conclusions that rest on a model that does not meet the criterion on residuals, the flow-rate study without its set-point run, and the residual DNA offset, which was recorded and not investigated. Five concern records the plan required that are not in the study record. These are the load results for aggregate and host cell protein, residual DNA and leached Protein A on the screening runs, the flow-through, wash and strip fractions with the mass balance, the packing tests of each campaign, and the feed lots. The other two are the fractionated runs of S-CEX-FRC, which no version of the plan foresaw, and the approaches this report takes where the plan sets no threshold, with capability given for the drug substance and not for the step. The last corrects an error of Version 1.0 of this report, which named the acetate load/wash buffer of screening run 6 a Tris buffer.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody developed on the Novacyte platform (CMC Biotech Working Group 2009). Its drug substance is purified by Protein A chromatography, low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration/diafiltration. Cation exchange is Step 7 of this train. It receives the neutralized pool of the low-pH viral inactivation step and passes its eluate pool to anion exchange.

The step is operated in bind and elute mode on a resin with a sulfonate ligand. Below its isoelectric point the antibody carries a net positive charge and binds the ligand. After a wash at the conductivity of the load, the antibody is eluted by a step change to the elution buffer, whose pH weakens the binding. The pool is collected between a threshold on the ascending edge of the elution peak and the stop collect threshold on the descending edge, both read as absorbance at 280 nm. Aggregate binds more strongly than monomer and elutes later, so the step separates the two by where the pool is cut. Host cell protein, residual DNA and leached Protein A that do not bind, or bind only weakly, leave in the load flow-through and the wash.

No quality attribute is set at this step. Aggregate is formed in the production bioreactor and added to during the low-pH hold. No later step removes aggregate, and UF/DF adds a little within its NOR. The pool of this step is therefore judged against the drug substance criterion less the largest rise expected across UF/DF (Section 7.1). Host cell protein is cleared here and again at anion exchange. No viral clearance is claimed for cation exchange. Enveloped virus is claimed at low-pH inactivation, anion exchange and virus filtration, and small viruses at anion exchange and virus filtration (International Council for Harmonisation 2023a).

The process is being transferred from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI under the Process Transfer Plan PTP-001. It operates there at a production bioreactor scale of 15,000 L. Table 1.1 lists the documents this report depends on and the document it rolls up into.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope

Document ID	Title	Relationship
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-007	Process Characterization Plan (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.2 Objectives

PCP-007 §2 sets six objectives. Objectives 2 and 3 were amended at Version 2.0 of the plan to include fragments, and objective 6 was added.

1. "To quantify the relationship between each studied parameter and each response of the step, over ranges wider than the ranges in which the step will be operated."
2. "To define the multivariate region over which the fitted models predict that the aggregate, the host cell protein and the fragment content of the eluate pool all meet the criteria in Table 7.1" of the plan.
3. "To establish a proven acceptable range for each parameter against the aggregate, the host cell protein and the fragment content of the pool, both with the rest of the step at target and with the rest of the step varying within its normal operating ranges."
4. "To classify each parameter under SOP-4001, from the size of its demonstrated effect and from the capability with which the parameter is controlled."
5. "To state the ranges, in-process controls and monitoring this step contributes to the control strategy, and to justify them for Stage 2."
6. "To verify the platform worst cases of cation exchange on which the criteria upstream rely."

Section 12 states the outcome against each objective.

1.3 Regulatory and scientific basis

The study belongs to Stage 1 of process validation, in which the commercial process is defined from development knowledge and the studies that support it are recorded

(U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) (International Council for Harmonisation 2009) and ICH Q11 (International Council for Harmonisation 2012). Prior knowledge and the risk assessment RA-001 identified the parameters that may affect the quality attributes, and the designed experiments measured and bounded those effects. Criticality is treated as a continuum, as described in ICH Q8(R2), within the risk management principles of ICH Q9(R1) (International Council for Harmonisation 2023b). The process model, the attribute framework and the risk methodology come from the A-Mab case study (CMC Biotech Working Group 2009). The designs and their analysis follow standard response surface practice (Montgomery 2017).

The master plan PCMP-001 sets the common approach of the campaign, and PCP-007 applies it to this step. Where this report states a rule, it cites the section of the plan or the master plan that sets it. The controlled procedures and method validations that governed the work are listed in Table 1.2. The SOP and AMV numbers are placeholders.

Table 1.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
	SOP-2103	Buffer Preparation, Release and Expiry Control
	SOP-3015	Purity by Non-Reduced CE-SDS
Analytical method validation	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3221	Aggregate by SEC-HPLC (verification-qualified)

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0, effective 2026-09-30, is issued under two Quality Assurance (QA) investigations. INV-001 (Acceptance criteria for proven-acceptable-range analysis set after data review) was opened on 2026-08-03 after a review of PCMR-001 Version 1.0 for the Biologics License Application (BLA). It was closed on 2026-09-18 with the classification major. It found that the criteria used to judge each step in the proven acceptable range analysis had

been set after the data were seen and appeared in no approved plan (INV-001 finding 1). INV-002 (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0) judged the response-surface models of the outputs with a criterion against the first three model adequacy criteria of PCMP-001 §7.2. Its corrective action CA3, adopted on 2026-09-15, analyses every response-surface output by model selection.

This revision changes the report in seven ways.

1. The acceptance criteria. Version 1.0 judged the pool against an aggregate limit of 1.00 % and a host cell protein limit of 193 ng/mg. Each was the drug substance criterion carried back through the nominal clearance of the later steps in this campaign and then divided by a margin, 5.0 for aggregate and 3.2 for host cell protein (INV-001 finding 2). Version 2.0 judges the pool against the intermediate acceptance criteria of PCMP-001 §7.3, derived from the drug substance criteria and the platform worst cases of the later steps (Section 7.1). Table 1.3 compares the two versions. Two criteria in force differ from their a priori values. The host cell protein criterion is wider than its a priori value of 400 ng/mg, and Section 7.1 argues that change by its risk. The aggregate criterion is tighter than the drug substance criterion of 5.00 % by the rise that UF/DF may add (Section 7.1).
2. Fragments. Fragments are judged at this step for the first time, against the drug substance criterion, on results by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS) under the analytical method decision AMD-001 (Section 3.3).
3. The models. Every output of the response-surface design is analysed by model selection instead of the full quadratic model of the plan (discrepancy PD-007-01, Section 3.5).
4. The scale-down model. The model is qualified for each output by its offset from manufacturing scale, not by a test of equivalence, and every judgement of an output with a measured offset is made at manufacturing scale (Section 3.1).
5. The judgement. Each quality attribute is judged on the 95 % prediction interval of a single run, and the design space is drawn on the same statistic (Section 3.5).
6. The verification. The report verifies the platform worst cases of cation exchange on which the criteria upstream rely (Section 7.4).
7. The record. The execution window of each study is recorded (Section 3.2). The events of execution that Version 1.0 recorded are reported as study events under their Version 1.0 numbers, and the departures from the plan as discrepancies (Section 13). Version 1.0 of this report stated that the load, the flow-through and the wash of each run were assayed and that the column was released against SOP-2008. No result of those assays or of that release is in the record, and Version 2.0 records each gap as a discrepancy. Version 1.0 of this report also named the load/wash buffer of screening run 6 a Tris buffer. It was an acetate buffer (discrepancy PD-007-12).

Table 1.3: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, cation exchange.

Attribute	Version 1.0[a)]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Aggregates, HMW (%)	≤ 1.00	R1, margin 5.0	≤ 4.90	Carried back, tightened[b)] from ≤ 5.00
Host cell protein (ng/mg)	≤ 193	R1, margin 3.2	≤ 418	Carried back, widened[c)] from ≤ 400
Fragments, LMW (%)	not assessed[d)]	–	≤ 2.00	Drug substance

[a)] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b)] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[c)] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[d)] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.

A NOR-propagated range is the range of one parameter over which its criterion is met while the other parameters vary within their NORs. Under the Version 1.0 limits and the Version 1.0 analysis, these ranges ended inside the NOR for protein load (10–27.6 g/L resin), elution buffer pH (pH 5.8–6.05) and elution stop collect (0.5–1.12 AU) against aggregate, and for protein load (10–28.4 g/L resin) and load/wash conductivity (4.55–7 mS/cm) against host cell protein. The upper end of the protein load NOR, 32 g/L resin, therefore appeared unsupported. Under Version 2.0 every output meets its criterion across the NOR, and every proven acceptable range of aggregate and host cell protein is the range studied (Section 7). The NOR and the set-point of each parameter of this step are the ones the studies were planned against, and none was changed at Version 2.0.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The cation exchange step has been operated on the Novacyte humanized IgG1 platform for three earlier products, X-Mab, Y-Mab and Z-Mab, and throughout A-Mab clinical and engineering manufacture (CMC Biotech Working Group 2009). The resin chemistry, the buffer system and the mode of operation are unchanged for A-Mab. This experience establishes the mechanisms of the step and identified the parameters worth studying. It also supplies the platform worst cases from which the criteria are derived (Section 7.1 and Section 7.4).

Aggregate presents more positive charge to the ligand than monomer does, so it is retained more strongly and elutes on the tail of the elution peak. The clearance of aggregate depends on how well the two species are resolved and on where collection stops. At the load pH, host cell proteins with an isoelectric point below that pH carry a net negative charge and leave in the load flow-through. Those with an isoelectric point above it bind with the antibody. The weakly bound among them are displaced in the wash, and the rest are separated from the antibody in the elution. A part of the host cell protein is held in association with the bound antibody and elutes with it. DNA carries a strong negative charge and is repelled by the ligand, so its clearance is large and is not expected to depend on the parameters studied. Free leached Protein A does not bind under load conditions, but the leached Protein A in complex with the antibody binds with it, which limits the clearance to a modest factor. Fragments carry less positive charge than intact antibody and elute on the leading side of the main peak. Yield is lost at both edges of the pool, before collection starts and after it stops.

PCP-007 §4.1 draws four expectations from this experience, and the multivariate designs were built to test them.

1. A higher protein load occupies more of the ligand. The monomer and aggregate peaks broaden and overlap, and more host cell protein is held with the bound antibody, so aggregate and host cell protein in the pool are expected to rise and yield to fall as load rises.
2. A higher load/wash conductivity is expected to strip more of the weakly bound host cell protein in the wash and to have little effect on the aggregate separation. The ionic strength of the load and the wash does not change the relative charge of aggregate and monomer at the elution pH, and that difference in charge is what separates them.
3. A higher elution buffer pH weakens binding and brings monomer and aggregate off closer together, so it is expected to raise aggregate in the pool.

4. The stop collect point is an absorbance on the descending edge, so a lower setting stops collection later. Stopping later is expected to take in more monomer and the leading part of the aggregate peak, which raises both the yield and the aggregate of the pool, and to add eluate that carries few fragments.

The plan adds that a higher conductivity is expected to remove more host cell protein at a high load than at a low one, and that elution flow rate, within its range, is expected to change the sharpness of the peak too little to move the aggregate content of the pool. Curvature was expected in the effects of elution stop collect and elution buffer pH (PCP-007 §6.3). The stop collect moves the end of the pool along the falling edge of the peak. The elution buffer pH changes the charge of the antibody, and so the position and the width of the peak against the collection thresholds. Because the aggregate peak has a maximum, equal steps in either setting add unequal amounts of aggregate to the pool.

2.2 Quality attributes in scope

The drug substance register assigns no quality attribute to this step. Aggregate is formed in the production bioreactor, host cell protein and residual DNA enter with the harvested cell culture fluid, leached Protein A is released at capture, and fragments form in culture and during the harvest hold. Table 2.1 lists the five attributes the step clears, with their drug substance criteria and their criticality. Criticality was assigned with Tool #1 of the case study, which combines the impact of an attribute with the uncertainty in that impact (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes in scope at cation exchange, with their drug substance criteria and criticality.

Category	CQA	Drug substance criterion	Criticality	Tool #1 score
Size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12
Process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36
	Residual DNA	≤ 0.001 ng/dose	VL	6
	Leached Protein A	≤ 5 ppm	L-M	16

VL, very low. L, low. M, moderate. H, high. A paired code such as M-H means moderate to high.

Three of the five attributes are responses of the designs. Aggregate has the highest criticality of the five and is the attribute the step exists to control. Host cell protein is cleared here and at anion exchange, so the pool of this step must be one that anion exchange can bring to the drug substance criterion. Fragments were measured by nrCE-SDS in the response-surface design (Section 3.3). Residual DNA and leached Protein A were measured across the step so that their clearance is known. The plan does not treat them as responses of the designs, and it claims a parameter dependence for neither unless the data show one. The residual DNA criterion of ≤ 0.001

ng per dose is the acceptable range that the case study claims for A-Mab in its Table 2.17. It is set far below the 10 ng per dose recommended by the World Health Organization, which the case study cites (CMC Biotech Working Group 2009). Step yield is a process performance attribute. It has no acceptance criterion, and it is read together with aggregate because the monomer cut away with the aggregate is lost from the pool.

2.3 Risk-based prioritization and parameter selection

All process parameters of the step were carried into the risk ranking and filtering of RA-001 (International Council for Harmonisation 2023b). Each parameter was ranked for its potential impact on a quality attribute and for its potential to interact with other parameters, and a parameter with no data or rationale was ranked at the highest level (CMC Biotech Working Group 2009). The assignment produced three groups.

1. Protein load, load/wash conductivity, elution buffer pH and elution stop collect were assigned to the multivariate designs. The first, third and fourth move the aggregate peak against the stop point, and load acts with conductivity on the host cell protein that survives the wash, so their effects cannot be assumed to add.
2. Elution flow rate was assigned to a univariate assessment. It acts on the separation through the residence time of the antibody in the column and the rate of mass transfer into and out of the resin, and within its range these were expected to change the sharpness of the elution peak only a little.
3. Parameters fixed by equipment design or held to platform specification, such as bed height, packing quality and buffer composition, were assigned no new study. Their ranges rest on the platform knowledge and on the procedures in Table 1.2.

The range studied of each parameter is wider than its NOR at both ends (Table 4.1). The ranges were set from the control capability of the commercial equipment and from the range over which the platform has been operated, so that the study bounds the step beyond the region in which it runs (International Council for Harmonisation 2009). The elution buffer pH range stops short of incomplete elution at its low end and of the co-elution of aggregate with monomer at its high end.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model is a laboratory column of the commercial resin type, packed to the commercial bed height and run at the commercial linear velocities (SOP-1001, SOP-2010). Load is expressed in grams of antibody per litre of resin and every buffer volume in column volumes, so each parameter of the study has its commercial value.

The qualification study S-CEX-SDM ran from 2026-02-04 to 2026-02-13, before the first characterization run on 2026-02-16. It compared 3 scale-down runs at the set-point with 5 A-Mab batches at manufacturing scale. The outputs compared were step yield, pool aggregate, pool host cell protein, residual DNA and leached Protein A, together with plate count and peak asymmetry (PCP-007 §5.1).

The offset of an output is the mean of the scale-down runs minus the mean of the manufacturing batches, with its two-sided 90 % confidence interval (PCMP-001 §7.1). For an output modelled on the log scale the offset is taken on that scale. It is then the ratio of the scale-down to the manufacturing geometric mean, and a scale-down value is divided by it to give the manufacturing-scale value. The model is qualified for an output with a criterion when the conclusion over the NOR is the same with the offset at either end of its interval. An output with no criterion at this step has its offset recorded only. Table 3.1 gives the offsets.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, cation exchange.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Aggregates, HMW (%)	3	5	0.75[a)]	0.75[a)]	×0.996[a)]	×0.883 to ×1.122	Yes / Yes / Yes	Yes
Host cell protein (ng/mg)			116.6[a)]	130.0[a)]	×0.897[a)]	×0.626 to ×1.284	Yes / Yes / Yes	Yes
Step yield (%)			92.2	91.4	0.8	−1.3 to 2.9	no criterion[b)]	n.a.[b)]
Column efficiency (plates/m)			2,472	2,622	−150	−429 to 130	no criterion[b)]	n.a.[b)]
Peak asymmetry			1.16	1.15	0.01	−0.08 to 0.11	no criterion[b)]	n.a.[b)]
Residual DNA (pg/mg)			21.5	32.3	−10.8	−17.1 to −4.5	no criterion[b)]	n.a.[b)]
Leached Protein A (ppm)			0.33[a)]	0.46[a)]	×0.711[a)]	×0.416 to ×1.213	no criterion[b)]	n.a.[b)]

[a)] Modelled on the log scale: geometric means, and the offset as the ratio of the scale-down to the manufacturing geometric mean. The manufacturing-scale value is the scale-down value divided by the ratio.
[b)] The output has no acceptance criterion at this step, so no conclusion depends on its offset and qualification does not apply. The offset is recorded.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

The model is qualified for both outputs that carry a criterion. The aggregate ratio is 0.996 (90 % CI 0.883 to 1.122). The host cell protein ratio is 0.897 (0.626 to 1.284). That interval is wide, because it rests on 3 and 5 runs and the manufacturing batches ranged from 83.3 to 176 ng/mg. The qualification for host cell protein depends on the criterion it is judged against. With the offset at the unfavourable end of its interval, the highest single-run limit over the NOR is 414 ng/mg. That meets the criterion in force of 418 ng/mg and would not meet the a priori criterion of 400 ng/mg. Under the a priori criterion the model would therefore not be qualified for host cell protein, and Section 7.1 argues the change of criterion on which the qualification rests. Section 5.3.4 gives the sensitivity of this qualification to the choice among the models within 2 AICc units of the selected one.

Among the outputs without a criterion, only residual DNA differed between the scales at the level of the two-sample t-test of Version 1.0 (p = 0.016, against p = 0.285 at the lowest for the other outputs compared). The scale-down pools carried 10.8 pg/mg less residual DNA than the manufacturing pools (90 % CI 4.5 to 17.1 pg/mg less). Version 1.0 of the plan required the cause of such a difference to be established before characterization went on, and it was not (discrepancy PD-007-04, Section 13). Residual DNA has no criterion at this step, so no conclusion of this report depends on its offset.

The qualification did not compare fragments at both scales, so their offset is unknown, not zero. Fragments are judged as the scale-down model gives them (PCMP-001 §7.1). That rests on the assumption, stated in PCP-007 §12, that fragments separate in the

same way at both scales. The scale-down column keeps the bed height, the linear velocity and the collection thresholds of the commercial column, and its plate count and peak asymmetry did not differ from manufacturing scale at the level of the Version 1.0 test (Table 3.1).

The centre-point runs of both designs were made at the set-point of every factor. Their spread measures the reproducibility of the model under study conditions and gives the pure error against which lack of fit is tested (Section 5.1). The scale-down column reproduces the chromatography of the commercial column, but not the hold times, pool volumes or mixing of the commercial suite. The conclusions of this report apply to the commercial step through the qualification above, and they are to be confirmed at scale in Stage 2.

3.2 Operation

Each run followed the operation of SOP-2010. The neutralized low-pH pool was brought to the load conductivity of the run and applied to the equilibrated column up to the protein load of the run. The wash used the same conductivity, because the platform holds load and wash conductivity equal, and the two were therefore varied as one factor. The antibody was eluted by a step change to the elution buffer of the run, and the pool was collected from a threshold on the rising edge of the peak to the stop collect threshold on its falling edge, both read as OD280. Strip, sanitization and re-equilibration followed SOP-2008. Buffer composition, packing and resin storage were not varied.

PCP-007 §5.1 required plate count and asymmetry to be tested, and the column released against SOP-2008, before each campaign. Plate count and asymmetry are recorded for the 3 scale-down runs of the qualification only, with no acceptance limits. No packing result is recorded for the screening, response-surface and flow-rate campaigns (discrepancy PD-007-08). PCP-007 §6.2 also required the feed lot of every run to be recorded. The record does not name the feed lot of any run (discrepancy PD-007-11).

Table 3.2 gives the execution window of each study made on the scale-down model of this step. The record names an instrumented system for the response-surface study only, the temperature-controlled chromatography chamber EQ-CHR-118. It was in calibration for the whole study, which closed on 2026-03-27, before its next calibration fell due on 2026-05-30. Every characterization study of the step closed by 2026-03-27, before Version 1.0 of this report was approved.

Table 3.2: Execution windows of the studies made on the scale-down model of cation exchange.

Study	Activity	Start	End	Instrumented system
S-CEX-SDM	Scale-down model qualification	2026-02-04	2026-02-13	none named
S-CEX-SCR	Screening design	2026-02-16	2026-02-27	none named
S-CEX-RSM	Response-surface design	2026-03-02	2026-03-27	EQ-CHR-118
S-CEX-FRC	Fractionated run, and fragments by SEC and by nrCE-SDS on the same samples[a)]	2026-03-02	2026-03-13	none named
S-CEX-UNI	Univariate assessment of elution flow rate	2026-03-16	2026-03-20	none named
S-CEX-VER	Cation exchange runs for PCR-006[b)]	2026-09-24	2026-09-25	none named
S-VI-WC	Cation exchange runs for PCR-006[c)]	2026-09-26	2026-09-29	none named

[a)] In neither version of the plan (discrepancy PD-007-09).
[b)] Discrepancy PD-006-02 of the low-pH viral inactivation report. Its results belong to that report.
[c)] Discrepancy PD-006-03 of the low-pH viral inactivation report. Its results belong to that report.

The screening design ran before the method decision AMD-001 of 2026-03-20, and the response-surface design ran across it (Section 3.3). Two studies of the low-pH viral inactivation report made cation exchange runs on this scale-down model after INV-001 closed. They are S-CEX-VER (2026-09-24 to 2026-09-25, discrepancy PD-006-02) and S-VI-WC (2026-09-26 to 2026-09-29, discrepancy PD-006-03), and Section 7.4 states what this report needs from them.

3.3 Analytical methods

Every response was measured by a validated method, listed in Table 1.2 and Appendix D. Pool aggregate was measured by size exclusion chromatography (SEC, AMV-3011) and reported as the percentage of high molecular weight species. Host cell protein was measured by enzyme-linked immunosorbent assay (ELISA, AMV-3012) and reported per milligram of antibody, so that the value does not move with the concentration of the pool. Residual DNA was measured by quantitative polymerase chain reaction (qPCR, AMV-3014) and leached Protein A by ELISA (AMV-3016). Step yield was calculated from the antibody mass in the load and in the pool, each from its absorbance at 280 nm. Appendix D gives the performance of each method.

Fragments are judged on nrCE-SDS results only (AMV-3020, under SOP-3015). From 2026-03-20 every study that measures fragments, at every step, measures them by nrCE-SDS, so that one method spans the train (AMD-001, PCMP-001 §7.8). Table 3.3 shows which method measured fragments in which design of this step. The screening design ran before the decision and has SEC fragment results only. Its samples were not retained for a retest by nrCE-SDS, and its SEC fragment results are kept in the record and are not used for any decision. The response-surface design measured fragments by both methods on every run. The flow-rate study measured no fragments, because Version 1.0 of the plan listed none for it.

Table 3.3: Methods used for size variants in the designed studies, cation exchange.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW		SEC
Fragments, LMW	SEC	SEC, nrCE-SDS[a)]

[a)] Both methods on the same samples of this design.

The decision followed the comparability study S-CEX-FRC (2026-03-02 to 2026-03-13), in which SEC and nrCE-SDS were run on the same samples (Table 3.4). The samples included the fractions F2 to F5 of a run of this step with the eluate collected in fractions, which were collected to read the elution profile against the stop collect. No version of the plan foresaw this study (discrepancy PD-007-09). SEC read fragments high wherever the sample still carried host cell protein, because the part of the host cell protein that elutes in the low molecular weight window of SEC is counted as fragment. In the Protein A pool, the viral inactivation pool and the cation exchange load it was 0.32 to 0.59 percentage points. From the cation exchange pool onward the two methods agreed to within 0.13 points (INV-001 finding 6). SEC remains the method for aggregate, where no such interference applies.

Table 3.4: Fragments by SEC and by non-reduced CE-SDS on the same samples (study S-CEX-FRC).

Sample	Replicate	Fragments, LMW (%)		SEC minus nrCE-SDS
		SEC	nrCE-SDS	
Protein A pool	1	1.32	0.90	+0.42
	2	1.42	1.10	+0.32
	3	1.32	0.82	+0.50
	4	1.29	0.83	+0.46
VI pool	1	1.43	0.84	+0.59
	2	1.39	0.88	+0.51
	3	1.42	0.88	+0.54
	4	1.49	1.02	+0.47
CEX load	1	1.36	0.85	+0.52
	2	1.52	1.03	+0.49
CEX fraction F2	1	1.65	1.56	+0.09
CEX fraction F3	1	0.64	0.76	-0.11
CEX fraction F4	1	0.40	0.49	-0.09
CEX fraction F5	1	0.42	0.32	+0.10
CEX pool	1	0.61	0.62	-0.01
	2	0.70	0.57	+0.13
	3	0.51	0.61	-0.10
	4	0.70	0.72	-0.02
AEX pool	1	0.55	0.65	-0.10
	2	0.68	0.70	-0.02

nrCE-SDS reports the sum of clipped species. The partial molecules seen under SDS (HHL, HH, HL, free light chain), which come from antibody lacking one or more interchain disulfide bonds, are excluded. This makes the two methods comparable.

Assay variability is part of the residual variance of each model and of the centre-point spread. A response measured by a less precise assay therefore carries a wider prediction interval and a less sensitive test of lack of fit. The ELISA for host cell protein has a precision of 9.5 % relative standard deviation (RSD), and the centre points show host cell protein as the least reproducible response (Section 5.1).

3.4 Sampling plan

PCP-007 §5.4 sampled each run at four points. The load after conditioning was to give the aggregate, host cell protein, residual DNA and leached Protein A entering the step. The flow-through and the wash were each to be sampled as one pooled fraction, to close the host cell protein mass balance. The eluate pool was to carry every response, and a strip fraction was to be retained and tested only where a run failed to close its mass balance.

The record of the designs holds the pool results of every run. It holds residual DNA and leached Protein A in the load and the pool of every response-surface run and of every run of the flow-rate study, and in the pools of the qualification runs. Three parts of the sampling plan are not in the record.

1. No load result for aggregate or host cell protein is recorded for any run (discrepancy PD-007-05). The clearance factors for aggregate and host cell protein in this report are therefore calculated against the level that the nominal batch of the process model brings to the step, as the verification rule of PCP-007 §7.6 specifies, and not against a measured load.
2. Residual DNA and leached Protein A were not measured on the screening runs (discrepancy PD-007-06).
3. No result for the flow-through, wash or strip fractions is recorded, so no mass balance was made and the strip test could not be triggered (discrepancy PD-007-07).

Version 1.0 of this report stated that the load, the flow-through and the wash were assayed, with no result recorded.

Replication is concentrated at the centre point. The screening design carries 3 centre-point runs and the response-surface design carries 4. The record holds the order of execution of every run. The centre points ran at positions 2, 9 and 11 of 19 in the screening design and at positions 10, 21, 24 and 25 of 28 in the response-surface design.

3.5 Statistical methods

All factors were analysed in coded units. Each factor is coded so that its set-point is 0 and the ends of its range studied are −1 and +1. At this step the set-point of every factor is the centre of its range, so the design centre and the set-point coincide. The letter codes of Table 4.2 are used for the model terms.

The screening design was analysed by ordinary least squares with all main effects and all two-factor interactions. An effect is twice the coded coefficient, the change in the response across the whole range of the factor, and a term is significant at $p < 0.05$ (PCMP-001 §6.2). The screening model identifies effects. It is not the predictive model of the step, because a model without quadratic terms cannot describe curvature.

The response-surface model is the predictive model. Every judgement over the NOR, every proven acceptable range, the design space and the verification of the platform worst cases come from it. The plan prescribed the full quadratic model of each output, with 14 terms besides the intercept (PCP-007 §5.5). Each output was instead analysed by the model selection of INV-002 corrective action CA3, in three steps (discrepancy PD-007-01).

1. The scale was chosen from the Box-Cox profile of the full quadratic (Box and Cox 1964). The log scale is used where the 95 % interval of the Box-Cox exponent λ contains 0 and excludes 1, and the raw scale otherwise.

2. On that scale, every hierarchical submodel of the full quadratic was fitted, 1,337 candidates for four factors, and the one with the lowest corrected Akaike information criterion (AICc) was taken. A hierarchical model keeps the main effects of any interaction or square it contains.
3. The selected model was judged against the four adequacy criteria of PCMP-001 §7.2. An output whose selected model fails one of the first three criteria has no adequate model. It is judged on its measured results and sets no range.

AICc is the small-sample form of Akaike's criterion (Hurvich and Tsai 1989), and selecting a model by such a criterion is standard practice (Burnham and Anderson 2002). It estimates how well a model would predict new runs. It rewards a close fit and penalizes every estimated term, and the residual variance is counted as one of those terms. Its correction for small samples raises the penalty when the runs are few for the number of terms, as they are in a design of 28 runs, so that the criterion does not favour too many terms. The search is exhaustive because a stepwise search changes one term at a time and stops when no single change lowers AICc, which can leave it short of a model that only a combination of changes reaches. On the data of this step a stepwise search, backward or forward, would have reached the selected model for aggregate. For host cell protein a backward search would have stopped at a model of 8 terms and a forward search at one of 3, with AICc values of −27.4 and −27.9 against −31.2 for the selected model, and for fragments both would have stopped at a model with a higher AICc. In this report the terms of a model are counted without the intercept.

The Box-Cox profile chose the log scale for aggregate (λ = 0.18, 95 % interval −0.40 to 0.62) and for host cell protein (λ = −0.01, interval −0.88 to 0.87). It kept the raw scale for fragments, whose interval contains 1. On the log scale the model is $\ln y = b_0 + b_1 x_1 + \cdots + \varepsilon$. Transformed back, $y = e^{b_0}\, e^{b_1 x_1} \cdots e^{\varepsilon}$, so each parameter multiplies the output by a fixed factor e^{b} per coded unit, and by e^{2b} from the low to the high end of its range. The scatter between runs multiplies the output too, so its size in the unit of the output grows with the level. This is the behaviour expected of a purification step, which removes a share of what is present, so that a parameter changes the share rather than a fixed amount. It is also expected of a method whose precision is a percentage of the level, as the ELISA for host cell protein is.

Least squares, and the t-tests, F-tests and prediction intervals built on it, assume residuals that are independent and normally distributed, with mean zero and the same variance over the whole range studied. For these two outputs the evidence for the log scale is the Box-Cox profile. The residuals themselves do not show the scatter growing with the level on the raw scale. For the full quadratic on the raw scale, the residual SD in the upper half of the predictions is only 5 % larger than in the lower half for aggregate and 6 % larger for host cell protein. For the full quadratic on the log scale it is smaller in the upper half (0.040 against 0.062 for aggregate). At 28 runs neither difference is large enough to decide the scale. The regression of the absolute residuals on the predicted values, one of the tests of the fourth criterion, finds no trend in the selected log-scale model of either output (Table 5.5).

On the log scale the criteria stay in their own units, % and ng/mg. Effects are fold changes, and Section 5.3 gives them. A prediction interval is calculated on the log

scale and transformed back, which keeps its 95 % coverage, because the probability that ln y lies between two values is the probability that y lies between their antilogs. The interval is asymmetric. At the least favourable NOR combination for host cell protein the median is 227 ng/mg and the interval runs from 178 to 289 ng/mg, further above the median than below it. The predicted value is the median of the output, its geometric mean, not its arithmetic mean. The upper prediction limit, which decides against an upper criterion, is not affected by that.

Each model is reported with R^2, adjusted R^2 and predicted R^2, the last from the prediction error sum of squares. The four adequacy criteria of PCMP-001 §7.2 are applied as PCP-007 §7.2 states them as tests.

1. The model F-test is significant at $\alpha = 0.05$.
2. Lack of fit against the pure error of the centre points is not significant at $\alpha = 0.05$.
3. Predicted R^2 falls short of R^2 by no more than 0.20. This is the limit on overfitting. A model that fits its own runs much better than it predicts runs left out of the fit has more terms than the data support.
4. The absolute residuals regressed on the predicted values, and the residuals regressed on the order of execution, show no slope significant at $\alpha = 0.05$. A run with a Cook’s distance above 4/n is examined by deleting it and repeating the judgement, and the model fails this criterion only where a run’s Cook’s distance exceeds the median of the F distribution with p and n − p degrees of freedom.

A model with no term has no F-test, so the first criterion does not apply to it. An output with a selected model that fails the fourth criterion is still used for the ranges and the design space, and that use is discrepancy PD-007-02. For each such model this report states the size of the trend against the criterion of its output, and it deletes each flagged run in turn, repeats the selection and judges the output over the NOR again. Where other models lie within 2 AICc units of the selected one, they are reported as a sensitivity, and the selected model decides.

A quality attribute is judged on a single batch. The judgement over the NOR asks whether the exact 95 % prediction interval of a single run, at manufacturing scale, lies within the criterion at every one of the 81 combinations of NOR low, set-point and NOR high of the four factors (PCP-007 §7.4). The interval is two-sided, so its upper end is a one-sided 97.5 % limit, and since every criterion at this step is an upper limit, the upper end decides. The design space is the set of settings within the ranges studied at which every output with a criterion meets it at both ends of the same interval. It is evaluated on a grid of 11 levels per factor, 14,641 points in all.

Two proven acceptable ranges are computed for each parameter and each output with a criterion and an adequate model (PCMP-001 §6.4). The set-point analysis scans the parameter across its range with the other factors at their set-points and judges the mean prediction, which is the median for a log-scale model, at manufacturing scale. The NOR-propagated analysis scans the parameter at 81 points and draws, at each, 2,000 settings of the other factors, each from a normal distribution centred on its set-point, with a standard deviation equal to the width of its NOR divided by 6, and clipped to the range studied. Each draw takes its own fitted coefficients from their sampling distribution and its own residual. A setting is accepted where the 97.5th percentile

of the simulated batches, at manufacturing scale, does not exceed the criterion. The range reported is the contiguous accepted interval that contains the set-point. The simulation uses the fixed seed 20240724. The NOR-propagated range is the one that supports a claim.

The verification of the platform worst cases follows PCP-007 §7.6. The clearance factor is the level that the nominal batch of the process model brings to the step divided by the pool level that the response-surface model predicts. For aggregate and host cell protein the prediction is first converted to manufacturing scale with the offset at its estimate, as PCMP-001 §7.1 rule 2 requires of every judgement against a criterion. Fragments have no measured offset and enter as the scale-down model gives them. The model gives the lowest clearance factor over the 81 combinations of the NOR and the one-sided 95 % prediction bound of a single batch at the least favourable combination. Capability is estimated for the drug substance from the integrated model of the train, which simulates 2,000 batches with every parameter drawn within its NOR (PCMP-001 §6.5), and is judged against the target capability index of 1.33 and the minimum of 1.0 (PCMP-001 §7.9). PCP-007 §5.5 also asked for the capability of the step, from the fitted models over draws within the NOR. It was not estimated (discrepancy PD-007-10).

3.5.1 Decisions for which the plan sets no threshold

PCP-007 §7 lists six decisions of Version 1.0 for which no numerical threshold was set, and PCP-007 §8 leaves a seventh to this report. The approach this report takes for each is as follows. Three of the decisions could not be made as planned, because the records they need are missing. The approaches and those three gaps are recorded as discrepancy PD-007-10.

1. A parameter dependence of residual DNA or leached Protein A “unless the data show one” (PCP-007 §4.2). In the response-surface design the clearance factor of residual DNA was analysed by the same model selection as every output, and a dependence would be shown by a selected model with a term that meets PCMP-001 §7.2. Leached Protein A could not be analysed in that way, because 6 of its 28 pool results lie below the limit of quantitation (LOQ) and a model of the others leaves them out. In the flow-rate study a dependence is judged by the difference between the results at the two ends of the range, with its 95 % confidence interval, as in item 5.
2. The repacking of a column that fails the packing tests (PCP-007 §5.1). The record holds plate count and peak asymmetry for the qualification runs only, with no limits, so no repacking decision can be reviewed (PD-007-08).
3. The attribution to the assay of a predicted value within assay variability of its criterion (PCP-007 §5.3). The distance of a prediction from its criterion is expressed in standard deviations (SD) of the precision of its method at that level.
4. The test of the strip fraction where a run fails to close its mass balance (PCP-007 §5.4). The record holds no flow-through, wash or strip result, so no mass balance was made and the test could not be triggered (PD-007-07).

5. The move of elution flow rate into a multivariate study after an effect “comparable in size with the effects estimated in the screening study” (PCP-007 §6.4). For each response of the flow-rate study, the difference between the mean results at the two ends of its range is given with its 95 % confidence interval (CI). The CI uses the residual SD of the same response in the response-surface design, with that design’s degrees of freedom, and the difference is set against the smallest effect found significant for the response in the screening design.
6. The identification of runs made on a lot “out of trend” (PCP-007 §12). The record names no feed lot for any run of the designs (PD-007-11), and the load results that would show a lot out of trend are not recorded either (PD-007-05). This decision could therefore not be made.
7. The judgement of the flow-rate range against the criteria (PCP-007 §8). The range is supported for an attribute when every individual result at both ends, at manufacturing scale, meets its criterion. It is not supported for an attribute the study did not measure.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the five parameters with their set-points, NORs, ranges studied, classes and study types. The range studied is the knowledge space of the step. The NOR is the range in which routine manufacture is controlled, and the classes are the outcome of Section 9.

Table 4.1: Parameters of the cation exchange step, with the NOR, the range studied and the class.

Parameter	Unit	Set-point	NOR	Range studied	Class	Study type
Protein load	g/L resin	25	18–32	10–40	WC-CPP	Multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	Multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	Multivariate
Elution stop collect	AU (OD280, descending)	1	0.7–1.3	0.5–1.5	WC-CPP	Multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	Univariate

The four multivariate parameters enter both designs as the coded factors of Table 4.2.

Table 4.2: Coded factors of the screening and response-surface designs.

Code	Factor	Unit
A	Protein load	g/L resin
B	Load/Wash conductivity	mS/cm
C	Elution buffer pH	pH
D	Elution stop collect	AU (OD280, descending)

4.2 Screening design

The screening study S-CEX-SCR ran from 2026-02-16 to 2026-02-27. It is a two-level full factorial in the four factors, 16 runs, with 3 centre-point runs, 19 runs in all. A full factorial was chosen over a fraction because protein load was expected to interact with each of the other three factors. It estimates every main effect and every two-factor interaction independently. The runs were executed in a randomized order, with

the centre points at the positions given in Section 3.4. Appendix A gives the design and its results.

4.3 Response-surface design

The response-surface study S-CEX-RSM ran from 2026-03-02 to 2026-03-27. It is a face-centred central composite design in the same four factors, with 16 factorial runs, 8 axial runs and 4 centre-point runs, 28 runs in all. The axial runs sit on the faces of the cube, so no run lies outside the range studied of any factor. The design adds the quadratic terms that the screening design cannot estimate. All four screening factors were carried into it. The design therefore did not need the augmentation that PCP-007 §6.3 requires when an active screening factor is left out of the response-surface design. Appendix B gives the design and its results.

4.4 Univariate assessment

The flow-rate study S-CEX-UNI ran from 2026-03-16 to 2026-03-20, within the window of the response-surface study. Elution flow rate was run at each end of its range, 100 cm/hr and 300 cm/hr, in duplicate, with every other parameter at its set-point. The plan also required a run at the set-point, which was not made, and the qualification runs at the set-point serve as the reference instead (discrepancy PD-007-03). Pool aggregate, pool host cell protein, step yield, and residual DNA and leached Protein A in the load and the pool were measured on each run. A study of one factor at a time cannot show whether flow rate interacts with the other parameters. That limit was accepted because flow rate sets the residence time in the resin, and with it the sharpness of the peak, while it changes neither the charge of the antibody nor the ionic strength that holds it to the ligand.

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean, SD and coefficient of variation of the centre points of both designs. These runs measure the reproducibility of the scale-down model under study conditions, and their spread is the pure error of the test for lack of fit.

Table 5.1: Centre-point reproducibility of the two designs.

Design	Response	Runs	Mean	SD	CV (%)
Screening	Aggregates, HMW (%)	3	0.78	0.08	10.2
	Host cell protein (ng/mg)	3	155	27.2	17.6
	Step yield (%)	3	92.5	1.31	1.4
Response surface	Aggregates, HMW (%)	4	0.77	0.04	5.6
	Host cell protein (ng/mg)	4	134	18.7	13.9
	Step yield (%)	4	91.8	2.89	3.2

CV, coefficient of variation (SD divided by the mean).

Host cell protein was the least reproducible response in both designs, with a coefficient of variation of 17.6 % in the screening design and 13.9 % in the response-surface design. This agrees with the expectation of PCP-007 §5.3. Aggregate varied by 10.2 % and 5.6 %, and step yield by 1.4 % and 3.2 %. The pure error has 2 degrees of freedom in the screening design and 3 in the response-surface design, so the tests for lack of fit are of modest power.

5.2 Factor effects in the screening design

Table 5.2 summarises the screening models. Each model is significant, and none shows significant lack of fit. With 2 degrees of freedom for pure error that test is weak, and the models are read for the effects they identify, not for prediction.

Table 5.2: Screening models: main effects and two-factor interactions.

Response	R^2	Adj. R^2	Pred. R^2	Model p	Lack of fit p	Residual SD
Aggregates, HMW (%)	0.97	0.94	0.87	< 0.001	0.439	0.10
Host cell protein (ng/mg)	0.94	0.86	0.47	< 0.001	0.380	35.4
Step yield (%)	0.92	0.83	0.50	0.002	0.694	1.15

Residual SD in the unit of the response. Adj., adjusted. Pred., predicted.

Aggregate was impacted by protein load, elution buffer pH and elution stop collect. Across the range studied, load raised pool aggregate by 0.60 percentage points ($p < 0.001$), pH by 0.34 points ($p < 0.001$) and stop collect by 0.32 points ($p < 0.001$). Significant interactions were found between load and pH (0.20 points), load and stop collect (0.15 points) and pH and stop collect (0.13 points). Load/wash conductivity had no significant effect on aggregate ($p = 0.710$).

Host cell protein was impacted by load/wash conductivity and protein load. Across the range, conductivity lowered pool host cell protein by 127 ng/mg ($p < 0.001$) and load raised it by 116 ng/mg ($p < 0.001$), with a significant negative interaction between them (−56.7 ng/mg, $p = 0.012$). On the raw scale, two effects that are additive on the log scale appear as an interaction, and with one effect positive and the other negative that interaction is negative. The response-surface analysis on the log scale examines whether a load × conductivity term remains (Section 5.3). Step yield fell with protein load by 5.18 percentage points across the range ($p < 0.001$), and no other term was significant. Appendix C gives every effect.

The 16 factorial runs of the screening design repeat the corners of the response-surface design. Over those corners the host cell protein results of the two designs did not differ systematically, with a geometric mean ratio of 0.98 (screening to response surface, $p = 0.739$). The largest difference was at the corner of protein load 40 g/L resin, load/wash conductivity 3 mS/cm, elution buffer pH 5.8 and elution stop collect 0.5 AU. There, screening run 9 gave 478 ng/mg at manufacturing scale and response-surface run 9 gave 265 ng/mg, a factor of 1.8. The screening result is the one result of either design above its criterion of 418 ng/mg. It lies above the upper 95 % single-run limit that the response-surface model gives at that corner, 368 ng/mg. The corner lies outside the NOR of all four parameters. On the log scale the two runs differ by 0.59, against 0.13 for one standard deviation of the difference between two single ELISA results, so the precision of the method does not explain the difference. The record gives no cause for it, and this report offers none. The response-surface model stays the predictive model, and Section 6 states what the result means for the design space.

5.3 Response-surface models

Table 5.3 gives the selected model of each output of the response-surface design against the adequacy criteria of PCMP-001 §7.2. The models of aggregate and host

cell protein meet the first three criteria. The aggregate model also meets the fourth, and the host cell protein model does not. The selected fragment model fails the limit on overfitting, so fragments have no adequate model. Step yield has no criterion and is reported for information.

Table 5.3: Selected response-surface models against the adequacy criteria of PCMP-001 §7.2.

Output	Scale	Terms	R^2	Adj. R^2	Pred. R^2	Gap	Model p	Lack of fit p	Crite-rion 4	Ade-quate
Aggregates, HMW (%)	log	8	0.97	0.96	0.94	0.035	< 0.001	0.467	met	Yes
Host cell protein (ng/mg)	log	6	0.95	0.94	0.92	0.031	< 0.001	0.846	not met	Yes
Fragments, LMW (%)	raw	3	0.48	0.41	0.27	0.211	0.001	0.712	met	No
Step yield (%)	raw	3	0.72	0.69	0.62	0.105	< 0.001	0.972	met	Yes

Terms, without the intercept. Adj., adjusted. Pred., predicted. Gap, R^2 minus predicted R^2, at most 0.20. Criterion 4, the residuals and influential runs. Adequate, the first three criteria met. Step yield has no criterion.

The plan's full quadratic model was adequate on the planned runs for aggregate and host cell protein, and the selection removed terms the data do not support (Table 5.4). For fragments the full quadratic failed the limit on overfitting by a wide margin. INV-002 assigned that model a refit, because no augmentation within the size of the planned design reached the required probability of success. Under corrective action CA3 the fragment model was selected instead, and the selected model of 3 terms also fails the limit. This step was not augmented.

Table 5.4: Re-evaluation of the response-surface models.

Attribute	Planned runs, full quadratic				Model used (terms)
	Runs	R^2	Pred. R^2	Gap	
Aggregates, HMW (%)	28	0.98	0.82	0.15	8 terms, log scale
Host cell protein (ng/mg)	28	0.97	0.83	0.14	6 terms, log scale
Fragments, LMW (%)	28	0.76	−0.40	1.16	None[a)]

[a)] No adequate model. The output is judged on its measured results. Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, on one block of runs (INV-002).

Table 5.5 gives the tests of the fourth criterion for each selected model. Only the run-order test of host cell protein falls below its level, and no run of any model has a Cook's distance above the median of $F(p, n - p)$.

Table 5.5: Residuals and influential runs of the selected models (fourth criterion of PCMP-001 §7.2).

Output	Spread p	Order p	Largest Cook's distance	Flag limit	Median of F(p, n − p)	Runs flagged	Criterion 4
Aggregates, HMW (%)	0.253	0.968	0.46 (run 7)	0.143	0.96	7, 10 and 15	met
Host cell protein (ng/mg)	0.242	0.043	0.24 (run 8)	0.143	0.94	8 and 13	not met
Fragments, LMW (%)	0.540	0.885	0.40 (run 4)	0.143	0.86	4, 5 and 8	met
Step yield (%)	0.724	0.949	0.50 (run 16)	0.143	0.86	16	met

Spread p, the regression of the absolute residuals on the predicted values. Order p, the regression of the residuals on the order of execution. Both are judged at $\alpha = 0.05$. Flag limit, 4/n. A flagged run is examined by deletion, and the model fails only for a run whose Cook's distance is above the median of F(p, n − p). Fragments have no adequate model and step yield has no criterion.

5.3.1 Aggregate

The aggregate model on the log scale has 8 terms, protein load, load/wash conductivity, elution buffer pH, elution stop collect, protein load × elution buffer pH, protein load × elution stop collect, elution buffer pH × elution stop collect and elution stop collect². It explains 97.3 % of the variation (adjusted R^2 0.962, predicted R^2 0.938). The model F-test is significant ($p < 0.001$), lack of fit is not ($p = 0.467$), and predicted R^2 falls short of R^2 by 0.035. The residuals show no trend against the predicted values ($p = 0.253$) or the order of execution ($p = 0.968$). Runs 7, 10 and 15 have a Cook's distance above 4/n = 0.143, the largest 0.46 for run 7, and none exceeds the median of F(p, n − p), 0.96.

From the low to the high end of its range, protein load multiplies pool aggregate by 1.85, elution buffer pH by 1.41 and elution stop collect by 1.37, each with the other factors at their set-points. There the interactions do not act, and a curvature term has the same value at both ends of a range, so these factors are the full change across the range. The conductivity term, kept by the selection, has $p = 0.055$ and a factor of 1.06, so no effect of conductivity on aggregate is demonstrated. The interactions make load act more strongly at a high pH and at a high setting of the stop collect. From the low to the high end of the load range, aggregate rises by a factor of 1.68 at pH 5.8 and by 2.03 at pH 6.2. The stop collect has a significant curvature, and with the other factors at their set-points aggregate is lowest near 0.63 AU. No curvature in pH was found. Table 5.6 gives the coefficients and Table 5.7 the analysis of variance.

Table 5.6: Coefficients of the selected models of aggregate and host cell protein (log scale, coded units).

Output	Term	Coefficient	SE	t	p
Aggregates, HMW (%)	Intercept	−0.2792	0.0204	−13.68	< 0.001
	A	0.3063	0.0152	20.13	< 0.001
	B	0.0311	0.0152	2.04	0.055
	C	0.1732	0.0152	11.39	< 0.001
	D	0.1566	0.0152	10.29	< 0.001
	AC	0.0482	0.0161	2.99	0.008
	AD	0.0408	0.0161	2.53	0.020
	CD	0.0392	0.0161	2.43	0.025
	D^2	0.1048	0.0255	4.12	< 0.001
Host cell protein (ng/mg)	Intercept	4.9606	0.0345	143.96	< 0.001
	A	0.3749	0.0257	14.60	< 0.001
	B	−0.3444	0.0257	−13.41	< 0.001
	C	0.0180	0.0257	0.70	0.491
	AC	−0.0511	0.0272	−1.88	0.075
	BC	−0.0850	0.0272	−3.12	0.005
	A^2	−0.0923	0.0430	−2.15	0.044

A, protein load. B, load/wash conductivity. C, elution buffer pH. D, elution stop collect. A letter pair is an interaction and a squared letter a curvature. Coefficients are on the natural log scale per coded unit.

Table 5.7: Analysis of variance of the selected models, with lack of fit against pure error (log scale).

Output	Source	Sum of squares	df	Mean square	F	p
Aggregates, HMW (%)	Model	2.8466	8	0.35583	85.40	< 0.001
	Residual	0.0792	19	0.00417	–	–
	Lack of fit	0.0693	16	0.00433	1.32	0.467
	Pure error	0.0099	3	0.00329	–	–
Host cell protein (ng/mg)	Model	4.8836	6	0.81393	68.55	< 0.001
	Residual	0.2493	21	0.01187	–	–
	Lack of fit	0.1876	18	0.01042	0.51	0.846
	Pure error	0.0618	3	0.02059	–	–

The residual is split into lack of fit and the pure error of the centre points.

Figure 5.1 shows the residuals of the aggregate model against the predicted values, on a normal probability plot and as measured against predicted values. The residuals scatter about zero with no trend that the spread test detects (Table 5.5), and they follow the normal line.

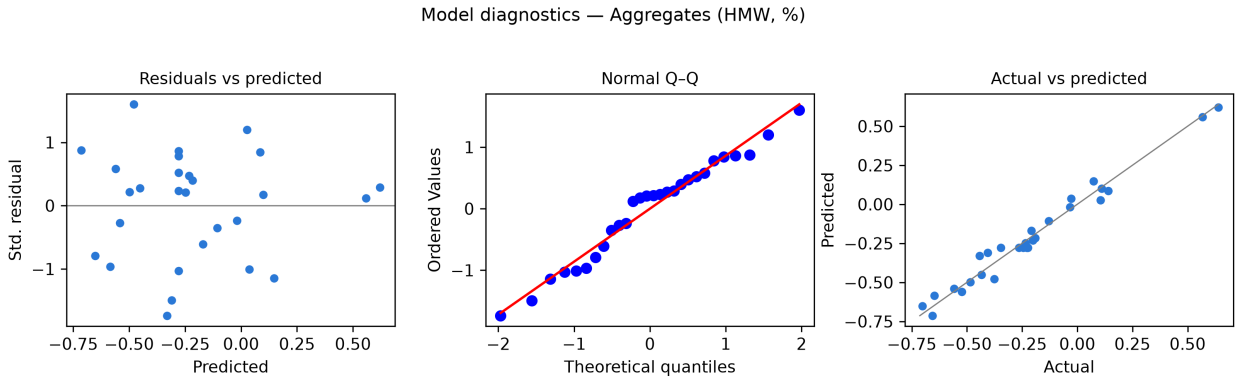


Figure 5.1: Diagnostics of the selected aggregate model (log scale), with the residuals against the predicted values, a normal probability plot and the measured against the predicted values.

5.3.2 Host cell protein

The host cell protein model on the log scale has 6 terms, protein load, load/wash conductivity, elution buffer pH, protein load × elution buffer pH, load/wash conductivity × elution buffer pH and protein load². It explains 95.1 % of the variation (adjusted R^2 0.938, predicted R^2 0.920). The model F-test is significant ($p < 0.001$), lack of fit is not ($p = 0.846$), and predicted R^2 falls short of R^2 by 0.031. The model does not meet the fourth criterion. Its residuals fall with the order of execution ($p = 0.043$), although they show no trend against the predicted values ($p = 0.242$). Runs 8 and 13 have a Cook's distance above 0.143, the largest 0.24 for run 8, and neither exceeds the median of $F(p, n - p)$, 0.94.

From the low to the high end of its range, with the other factors at their set-points, protein load multiplies pool host cell protein by 2.12 and load/wash conductivity multiplies it by 0.50. The two effects are of the same size within their standard errors. With the other factors at their set-points the model gives 99.7, 159 and 211 ng/mg at manufacturing scale for loads of 10 g/L resin, 25 g/L resin and 40 g/L resin, and 225 and 113 ng/mg at conductivities of 3 mS/cm and 7 mS/cm. The selected model keeps a negative curvature in load ($p = 0.044$). Its coefficient, −0.092 on the log scale, is smaller than the SD of the centre points on the same scale, 0.144, and 2 of the 5 other models within 2 AICc units leave it out (Section 5.3.4). PCP-007 §12 states that an effect smaller than the spread of the centre points will not be claimed, so the curvature is kept in the model for prediction and is not claimed as a finding. Elution buffer pH has no significant main effect ($p = 0.491$), but it changes the effect of conductivity ($p = 0.005$). From the low to the high end of the conductivity range, host cell protein falls by a factor of 0.60 at pH 5.8 and of 0.42 at pH 6.2. The load × pH term is kept with $p = 0.075$. On the log scale the full quadratic gives the load × conductivity term $p = 0.988$, so the interaction the screening design found on the raw scale is not found on the log scale.

Figure 5.2 shows the diagnostics of the host cell protein model.

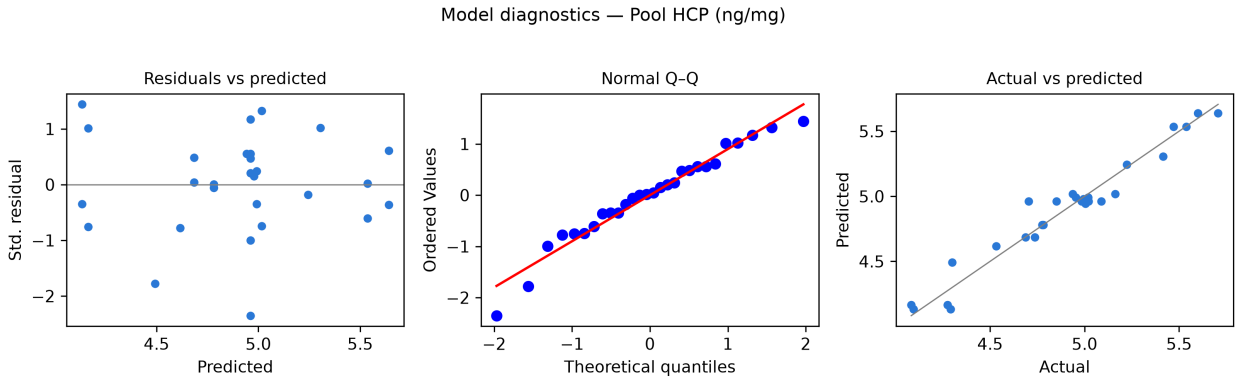


Figure 5.2: Diagnostics of the selected host cell protein model (log scale), with the residuals against the predicted values, a normal probability plot and the measured against the predicted values.

Figure 5.3 plots the residuals of the host cell protein model against the order of execution. The slope is −0.0045 per run on the log scale, so that over the 27 positions of the study the residuals fell by 0.121, a factor of 1.13 between the first and the last run at the same settings. The record gives no cause for the drift. Added in full to the highest single-run limit over the NOR, 289 ng/mg, it gives 326 ng/mg against 418 ng/mg. The conclusion over the NOR therefore does not depend on the trend. Outside the NOR it can. Added to the highest single-run limit over the whole region studied, 408 ng/mg, the drift gives 461 ng/mg, and 1.13 % of the grid would then exceed the criterion. Added to the highest limit over the NOR with the offset at the unfavourable end of its interval, 414 ng/mg, it gives 467 ng/mg, so with the drift the qualification for host cell protein would not hold. The edge of the design space and the qualification for host cell protein are therefore sensitive to the drift, whose cause is unknown.

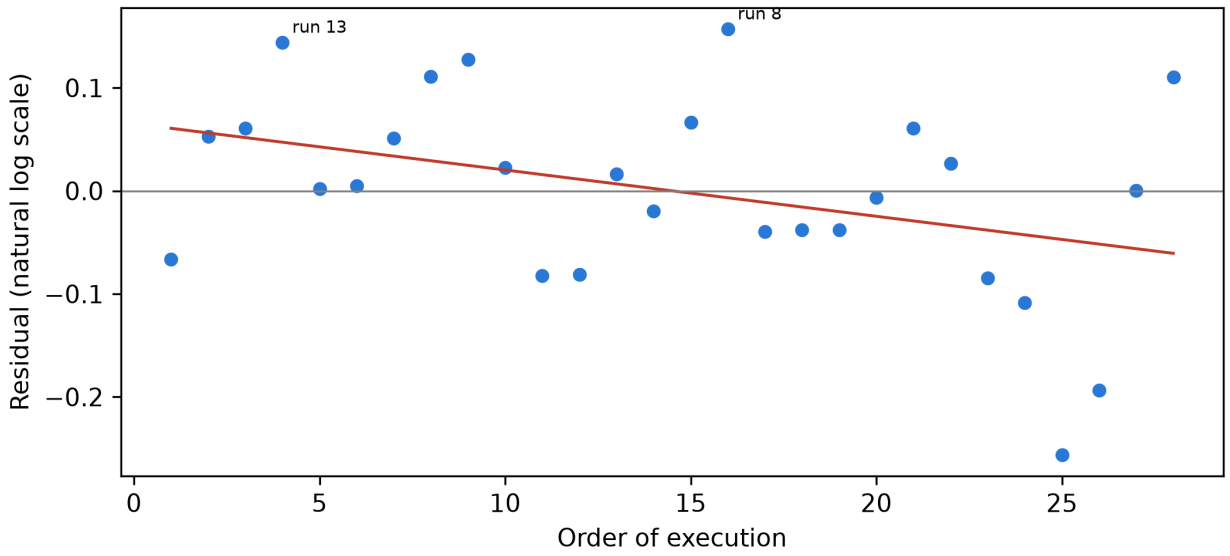


Figure 5.3: Residuals of the selected host cell protein model (log scale) against the order of execution, with the fitted straight line.

5.3.3 Influential runs

Each run with a Cook's distance above 4/n was deleted in turn, the model selection was repeated on the other 27 runs, and the output was judged over the NOR again (Table 5.8). With any one of these runs deleted, aggregate and host cell protein still meet their criteria over the NOR, the whole region studied still meets them on the single-run limit, and the verification of their platform worst cases gives the same outcome as in Section 7.4. The terms of the selected models do change. Every deletion changes the aggregate model, and deleting run 7 moves it to the raw scale. Deleting run 8 or run 13 moves the host cell protein model to the raw scale, where it keeps a load × conductivity interaction, as the screening analysis on the raw scale did. The choice of scale for host cell protein therefore rests on these two runs, and no conclusion over the NOR rests on that choice.

Table 5.8: Selection repeated with each flagged run deleted.

Output	Run deleted	Scale	Terms	Highest single-run limit over the NOR	Region inside (%)	Lowest clearance factor in the NOR	Bound of a single batch
Aggregates, HMW (%)	none	log	8	1.40	100	1.58	1.40
	7	raw	8	1.40	100	1.52	1.39
	10	log	6	1.37	100	1.62	1.43
	15	log	6	1.41	100	1.58	1.39
Host cell protein (ng/mg)	none	log	6	289	100	40.2	32.9
	8	raw	5	260	100	40.4	35.9
	13	raw	5	258	100	40.4	36.2

Limits at manufacturing scale, in the unit of the output. Region inside, the share of the grid over the region studied at which the single-run limit meets the criterion. Clearance factor and bound as in the verification of the platform worst cases (one-sided bound of a single batch), at manufacturing scale.

5.3.4 Models within two AICc units of the selected model

Other models lie within 2 AICc units of the selected one for both outputs (Table 5.9). For aggregate, the one other model differs only by leaving out the conductivity main effect, and it gives the same conclusions. For host cell protein, 5 other models differ in the load × pH term and in curvature in load and pH. All of them meet the first three adequacy criteria and give the same judgement over the NOR.

Three results differ among the host cell protein models. First, 4 of the 5 other models put a small part of the region studied outside the criterion on the single-run limit, at most 0.38 % of the grid. That part lies at the lowest conductivity and at loads of 37 g/L resin and above. Second, 3 models give a highest single-run limit over the NOR above the criterion of 418 ng/mg with the offset at the unfavourable end of its interval, up to 432 ng/mg. Under them the scale-down model would not be qualified for host cell protein. Third, 3 models give a lowest clearance factor within the NOR of 38.8 to 39.5, below the platform worst case of 40.0. Under any of them, rule 1 of PCMP-001 §7.5 would replace the platform value with the bound of a single batch and tighten the host cell protein criterion of the Protein A pool. They show that the qualification for host cell protein, the verification of its platform worst case and the edge of the design space at that corner all lie close to their limits.

Table 5.9: Models within two AICc units of the selected model.

Output	AICc above best	Terms added (+) or removed (−)	Gap	Limit over the NOR	Qualified	Region outside (%)	Clearance factor
Aggregates, HMW (%)	0.00	selected model	0.035	1.40	Yes	0.00	1.58
	0.62	−B	0.033	1.39	Yes	0.00	1.60
Host cell protein (ng/mg)	0.00	selected model	0.031	289	Yes	0.00	40.2
	0.36	−AC	0.032	298	No	0.15	39.5
	1.46	$+C^2$	0.035	293	No	0.00	39.5
	1.58	$-A^2$	0.027	282	Yes	0.15	41.8
	1.58	$-AC\ -A^2$	0.028	290	Yes	0.38	41.1
	1.82	$+C^2\ -AC$	0.036	302	No	0.15	38.8

Letters as in the coefficient table. Gap, R^2 minus predicted R^2. Limit over the NOR, the highest single-run limit at manufacturing scale. Qualified, the conclusion over the NOR the same with the offset at either end of its interval. Region outside, the share of the grid over the region studied at which the single-run limit exceeds the criterion. Clearance factor, the lowest within the NOR, as in the verification of the platform worst cases at manufacturing scale. Every model shown meets the first three adequacy criteria.

5.3.5 Fragments

Fragments have no adequate model. The selected model keeps elution buffer pH, elution stop collect and elution buffer pH × elution stop collect on the raw scale. Its F-test is significant ($p = 0.001$) and lack of fit is not ($p = 0.712$), but predicted R^2 falls short of R^2 by 0.211, above the limit of 0.20. The only other model within 2 AICc units fails the same limit. Fragments are therefore judged on their measured results, and no range and no predicted level is derived for them. Every one of the 28 nrCE-SDS

results of the response-surface design lay between 0.47 % and 0.76 %, below the criterion of 2.00 %, at settings that span the range studied of all four parameters.

The gap of the fragment model lies close to its limit. With run 4 or run 8 deleted the same terms would meet it. The stop collect term of the fragment model has the direction expected in Section 2.1, fragments rising as the stop collect setting rises and collection stops earlier, but no effect on fragments is claimed from a model that does not meet PCMP-001 §7.2.

5.3.6 Step yield

Step yield has no criterion. Its selected model on the raw scale keeps protein load, load/wash conductivity and protein load × load/wash conductivity and explains 72.4 % of the variation. With the other factors at their set-points the model gives 91.6 % in the scale-down model, and from 94.7 % to 88.6 % across the load range. The selected model has no stop collect term. The stop collect was also tested directly. Its term has p = 0.239 in the plan's full quadratic and its effect p = 0.268 in the screening model, so no effect of the stop collect on yield was detected. The selection is sensitive to one run. Run 14 is the run with the temperature excursion. With it deleted, the selection keeps protein load, load/wash conductivity and elution stop collect and drops the protein load × load/wash conductivity interaction. In that model the stop collect term is −0.76 percentage points per coded unit (p = 0.078), yield falling as the setting rises, which is the direction the mechanism expects (Section 13.1.2).

5.3.7 Response surfaces

Figure 5.4 shows the predicted surfaces of the three outputs over protein load and load/wash conductivity, and Figure 5.5 over elution buffer pH and elution stop collect, with the other two factors at their set-points. The surfaces are medians in the scale-down model for aggregate and host cell protein.

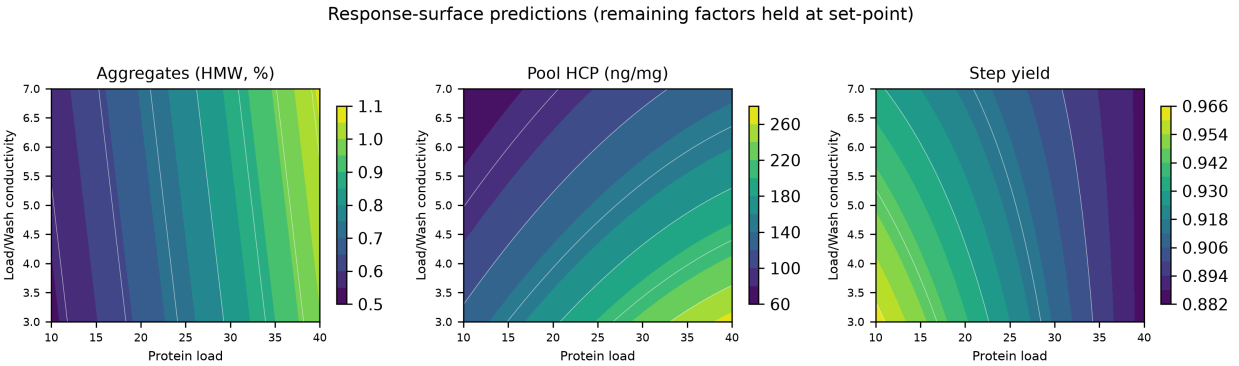


Figure 5.4: Predicted pool aggregate (%), host cell protein (ng/mg) and step yield (%, plotted as a fraction) over protein load and load/wash conductivity, with elution buffer pH and elution stop collect at their set-points (scale-down model).

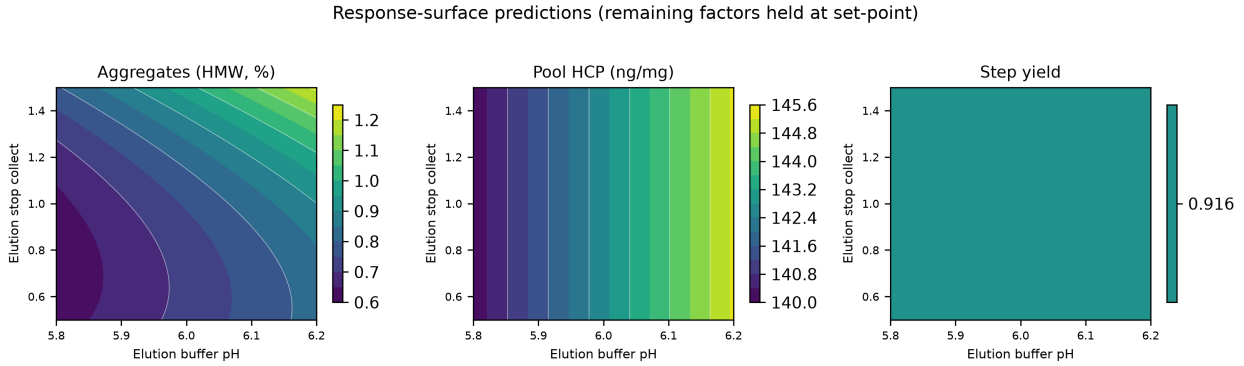


Figure 5.5: Predicted pool aggregate (%), host cell protein (ng/mg) and step yield (%, plotted as a fraction) over elution buffer pH and elution stop collect, with protein load and load/wash conductivity at their set-points (scale-down model).

5.4 Residual DNA and leached Protein A

Residual DNA was quantified in the load and the pool of all 28 response-surface runs. The load carried 558 to 898 pg/mg and the pool 12.7 to 43.6 pg/mg, so the step cleared 1.24 to 1.85 log10 (mean 1.53). The selected model of the clearance factor, on the log scale, has no term. Within the range studied, therefore, the clearance of residual DNA does not depend on any of the four parameters, at the precision of these runs and of the qPCR method (15 % RSD). The process model credits the step with 1.5 log10, within the range measured. Appendix B gives the result of every run.

Leached Protein A was quantified in the load of every run, at 3.15 to 3.97 ppm. In 22 pools it was quantified at 0.27 to 0.66 ppm, a clearance factor of 5.36 to 14.7. In the other 6 pools it lay below the LOQ of 0.25 ppm, so their clearance factors are known only to exceed 12.6 to 15.2. The model selection applied to the 22 quantified pools chooses the log scale and keeps protein load ($p = 0.042$), pool leached Protein A rising with load. That fit leaves out the 6 results below the LOQ, which were at loads of 10 g/L resin, 25 g/L resin and 40 g/L resin and include 3 at the highest load, so the data do not show a dependence on load, and none is claimed. The offset of leached Protein A in Table 3.1 is a ratio because that fit is on the log scale.

Every response-surface run cleared leached Protein A by more than the platform worst case of 3.16 on which the Protein A criterion relies. Leached Protein A is not a response of the designs, so no model replaces that platform value (Section 7.4).

5.5 Univariate assessment of elution flow rate

Table 5.10 gives the results of the flow-rate study at both ends of the range, with the qualification runs at the set-point as the reference. Aggregate and host cell protein are stated at manufacturing scale. Every result at both ends met its criterion, so

the range studied of 100–300 cm/hr is supported for aggregate and host cell protein by its own data. Fragments were not measured, so the study supports no range for fragments.

Table 5.10: Results of the flow-rate study (S-CEX-UNI), with the qualification runs at the set-point as the reference.

Flow rate (cm/hr)	Run	Aggregate (%)	HCP (ng/mg)	Yield (%)	DNA load (pg/mg)	DNA pool (pg/mg)	DNA clearance (log10)	Leached Protein A (ppm)
100	1	0.75	145	93.1	785	26.3	1.47	0.30
	2	0.69	131	93.6	638	12.3	1.71	0.34
300	1	0.68	145	91.9	551	43.2	1.11	< LOQ
	2	0.82	137	91.7	571	21.0	1.43	< LOQ
200[a)]	mean of 3	0.75	130	92.2	n.t.	21.5	n.t.	0.33

[a)] Scale-down runs of the qualification study S-CEX-SDM at the set-point, made before the characterization runs. Aggregate, host cell protein and leached Protein A as geometric means, as in the offsets. The load was not tested for residual DNA in these runs.
Aggregate and host cell protein at manufacturing scale, the scale-down result divided by the ratio offset. Yield, residual DNA and leached Protein A as measured in the scale-down model. LOQ of leached Protein A 0.25 ppm. n.t., not tested.

Table 5.11 gives, for each response, the difference between the two ends with its 95 % CI, to be set against the smallest effect found significant in the screening design (Section 3.5.1 item 5). No difference was detected in aggregate, host cell protein or step yield. For aggregate and host cell protein the differences between the ends, 0.03 percentage points and 2.69 ng/mg in the scale-down model, are smaller than the smallest effect found significant in the screening design, 0.13 points and 56.7 ng/mg. For step yield the interval of the difference, −5.19 to 2.00 points, is wide enough to include an effect as large as that of protein load in the screening design (−5.18 points). The study is too small to exclude such an effect on yield.

Table 5.11: Elution flow rate, the high end of the range against the low end.

Response	Comparison	High end against low	95 % CI	df	p	p, duplicates only
Aggregates, HMW (%)	ratio	1.04	0.91 to 1.19	19	0.538	0.730
Host cell protein (ng/mg)	ratio	1.02	0.82 to 1.28	21	0.838	0.741
Step yield (points)	difference	−1.59	−5.19 to 2.00	24	0.369	0.030
Residual DNA clearance (log10)	difference	−0.32	−0.64 to −0.01	27	0.044	0.252

Ratio, the geometric mean at the high end divided by that at the low end (log-scale models). Difference, the high end minus the low end. CI and p from the residual SD of the response in the response-surface design, with its df (for residual DNA, the SD of the clearance over the response-surface runs). p, duplicates only, from the scatter of the duplicates alone, 2 df.

Residual DNA clearance was 0.32 log10 lower at the high end than at the low end (95 % CI 0.01 to 0.64 log10 lower, p = 0.044). It is the only response of the study for which the comparison reaches p < 0.05. It rests on two runs at each end, and

the lowest clearance, 1.11 log10 at 300 cm/hr, lies below the range of the response-surface runs. Residual DNA was not measured in the screening design, so there is no screening effect to set this difference against. With the scatter of the duplicates alone, the DNA comparison would not reach significance ($p = 0.252$) and the yield comparison would ($p = 0.030$). That estimate of the scatter rests on two pairs of runs. It is larger than the residual SD of the response-surface design for aggregate and residual DNA clearance, and smaller for host cell protein and step yield. Both pools at the high end carried leached Protein A below the LOQ, so its clearance there is known only as a lower bound.

The study detected no effect of flow rate on aggregate or host cell protein comparable in size with the effects of the screening design. It detected none on yield, but it cannot exclude one as large as the effect of protein load. Flow rate was not carried into a multivariate study. Residual DNA clearance was significantly lower at the high end of the flow-rate range. Residual DNA has no criterion at this step, and Section 9 judges the practical relevance of the difference against the drug substance.

5.6 Mechanistic interpretation

Protein load and elution buffer pH acted on aggregate in the directions expected. A higher load occupies more of the ligand and broadens the elution peaks, so the aggregate front overlaps more of the monomer. A higher pH lowers the positive charge of both species, which weakens their binding and brings them off closer together. The positive load × pH interaction fits the same picture, since the separation that load erodes is the separation that a high pH has already compressed. Within this range conductivity has no demonstrated effect on aggregate, as expected. The ionic strength of the load and the wash weakens the binding of every species during loading and washing. It does not change the relative charge of aggregate and monomer at the elution pH, which is what separates them.

The elution stop collect acted on aggregate in the opposite direction to the expectation of PCP-007 §4.1. A higher setting stops collection earlier on the descending edge and should leave more of the aggregate tail out of the pool. In both designs aggregate instead rose as the setting rose, by 1.37-fold across the range in the response-surface model, with its lowest value near 0.63 AU. The selected yield model has no stop collect term, where a later stop was expected to recover more monomer (Section 5.3.6). The mechanism of Section 2.1 does not explain these results, and this report gives no cause for them.

Protein load and load/wash conductivity acted on host cell protein in the directions expected. More load leaves more host cell protein held with the bound antibody, and a higher ionic strength strips more of the weakly bound species in the wash. On the log scale each acts as a constant factor, so in ng/mg conductivity removes more host cell protein at a high load, where there is more to remove, as the plan expected. A change with load of the share that conductivity removes was not found. The pH × conductivity interaction was not expected, and the mechanism of Section 2.1 does not explain it.

Residual DNA clearance did not depend on any of the four factors of the response-surface design, as expected for a species repelled by the ligand. The lower clearance at the highest flow rate is not explained by that mechanism either. Leached Protein A was cleared by a modest factor, as expected for a species partly bound in complex with the antibody. Yield fell with load, as expected when broader peaks lose more monomer at their edges.

6 Design space

The design space of the step is the region of its four multivariate parameters, within their ranges studied, over which every output with a criterion meets it at both ends of the 95 % prediction interval of a single run (PCMP-001 §6.2, PCP-007 §7.5). Aggregate and host cell protein enter through their selected models, at manufacturing scale. Fragments have no adequate model and no measured offset. They constrain the region through their measured results in the scale-down model, all of which met the criterion. On the grid of 14,641 points, 100 % of the region meets every criterion in force. The region drawn on the mean prediction, for comparison, is also the whole region (100 %).

That result rests on the widening of the host cell protein criterion (Section 7.1). At the a priori criterion of 400 ng/mg, 99.85 % of the grid would meet every criterion. The 22 points outside lie at 40 g/L resin, 3 mS/cm and elution buffer pH 6.16 and above, with single-run limits up to 408 ng/mg.

No attribute bounds the region within the ranges studied. Host cell protein comes nearest to its criterion. Its highest single-run limit over the region is 408 ng/mg at protein load 40 g/L resin, load/wash conductivity 3 mS/cm and elution buffer pH 6.2, at any stop collect setting, 9.75 ng/mg below the criterion. The highest single-run limit of aggregate is 2.20 % at protein load 40 g/L resin, load/wash conductivity 7 mS/cm, elution buffer pH 6.2 and elution stop collect 1.5 AU, against 4.90 %. The NOR lies inside the ranges studied for every parameter, and inside the design space.

Four results bound this claim near the corner of highest load and lowest conductivity. Screening run 9 at such a corner gave host cell protein above the criterion (Section 5.2). Of the other host cell protein models within 2 AICc units of the selected one, 4 put a small part of that corner outside the criterion (Section 5.3.4). At that corner each design has a single run, and the two runs differ by a factor of 1.8. With the run-order drift of host cell protein added, 1.13 % of the grid would exceed the criterion (Section 5.3.2). The design space has not been confirmed at manufacturing scale anywhere near the edges of the ranges studied. The claim also holds only at the set-point of elution flow rate, which is not a factor of the designs, and only within the qualified life of the resin.

Under ICH Q8(R2), working within an approved design space does not need a regulatory change (International Council for Harmonisation 2009). Any planned movement is still assessed under the change management of the quality system (International Council for Harmonisation 2008). The NOR of this step was not changed, and the plan’s rule that a range is not narrowed to make a margin the data do not require applies (PCP-007 §7.5).

7 Proven acceptable ranges

7.1 Acceptance basis

Each output is judged against its intermediate acceptance criterion (PCMP-001 §7.3). The rule carries each drug substance criterion backward through the train, using the worst case that each later step showed on the platform for X-Mab, Y-Mab and Z-Mab, and applies no margin. Table 7.1 gives the derivation for this step. The derivation of the a priori criteria uses no result of the studies they judge.

Table 7.1: Intermediate acceptance criteria for the cation exchange output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Host cell protein (ng/mg)	≤ 100	4.18	0.00	≤ 418	Carried back, widened[a)]
Aggregates, HMW (%)	≤ 5.00	1.00	0.10	≤ 4.90	Carried back, tightened[b)]
Fragments, LMW (%)	≤ 2.00	1.00	0.00	≤ 2.00	DS criterion[c)]

[a)] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[b)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[c)] No downstream step changes the attribute, so the drug-substance criterion applies.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

No later step changes fragments in the derivation, so the pool is judged against the drug substance criterion of 2.00 % for fragments. No later step removes aggregate. PCMP-001 assumed that none adds any either, so the a priori aggregate criterion of the pool was the drug substance criterion of 5.00 %. UF/DF raises aggregate within its NOR. The UF/DF report PCR-010 gives the one-sided 95 % prediction bound of a single run for that rise, 0.104 percentage points, with the offset of its scale-down model applied. Carried back from the drug substance criterion, that bound tightens the criterion of this pool to 4.90 %. The tightening changes no conclusion of this

report, because the highest single-run limit of aggregate over the NOR, 1.40 %, meets both values. For host cell protein the drug substance criterion of 100 ng/mg is carried back through anion exchange. With the platform worst case of anion exchange, a clearance factor of 4.00, the criterion is 400 ng/mg a priori. The anion exchange report PCR-008 verified that worst case, and the one-sided 95 % prediction bound of a single batch at the least favourable combination of its NOR, 4.18, replaced it. The criterion in force is therefore 418 ng/mg.

That widening is argued here by its risk, in the four parts of PCMP-001 §7.5.

1. Who bears the risk. A wider criterion lets the pool of this step carry up to 418 ng/mg into anion exchange instead of 400 ng/mg. If anion exchange then cleared only its platform worst case, the drug substance could carry 104 ng/mg, above its criterion. The risk falls on the producer where such a result is detected, and on the patient only where it is not.
2. Detection. Host cell protein is a release test of the drug substance, by the ELISA AMV-3012 with an LOQ of 1.00 ng/mg. Its reagent covers 80 % of the host cell proteins seen by 2D western blot against the platform CHO host cell protein antigen, so an excursion made only of the species outside that coverage would not be detected.
3. What the later steps must clear. Anion exchange must clear 4.18-fold to bring the criterion in force to the drug substance criterion, which is its bound. The bound in force comes from the 28 planned runs of its response-surface design, on a model that meets the first three adequacy criteria (R^2 minus predicted R^2 0.024, lack of fit $p = 0.824$). On all 40 of its runs, with the 12 runs of the INV-002 augmentation, the bound is also 4.18. The bound holds within the NOR of anion exchange. At the edges of its range studied the lowest clearance factor measured was 3.55, below the platform value.
4. The drug substance. No simulated batch of the integrated model exceeded the host cell protein criterion of the drug substance, so the probability is below 0.15 % (Section 8).

The widening changes no conclusion of this report over the NOR. The highest single-run limit of the pool over the NOR, 289 ng/mg, meets the a priori criterion as well. The widening does decide two results outside that judgement. At the a priori criterion the scale-down model would not be qualified for host cell protein (Section 3.1), and the design space would cover 99.85 % of the region studied instead of all of it (Section 6). The proven acceptable ranges would not change.

Residual DNA, leached Protein A and step yield have no criterion at this step, so no range is derived from them. The platform history gives no worst-case DNA clearance for the steps after Protein A, so no DNA criterion can be carried back, and residual DNA is judged on the drug substance only (PCMP-001 §7.3). Leached Protein A is judged in the Protein A pool, against a criterion of 158 ppm carried back through the platform worst cases of cation exchange and anion exchange.

7.2 Outputs against their criteria

Table 7.2 gives each output against its criterion. Every output meets its criterion across the NOR.

Table 7.2: Cation exchange outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model, or measured results where it has none[a)b)]		95 % limits of a single run over the NOR[b)]	Met across the NOR[c)]
		At the set-point	Range over the NOR		
Aggregates, HMW (%)	≤ 4.90	0.76[d)]	0.56 to 1.21[d)]	0.49 to 1.40[d)]	Yes
Host cell protein (ng/mg)	≤ 418	159[d)]	110 to 227[d)]	86.5 to 289[d)]	Yes
Fragments, LMW (%)	≤ 2.00	n.a.[a)]	0.47 to 0.76[a)]	n.a.[a)]	Yes

[a)] No adequate model (PCMP-001 §7.2): the range of the measured results of the designs, less the offset of the scale-down model where one is known, on which the output is judged.
[b)] At manufacturing scale: the scale-down prediction minus the offset of the scale-down model for this output, or divided by it where the offset is a ratio. Fragments have no measured offset and are given as measured in the scale-down model.
[c)] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty).
[d)] Modelled on the log scale: the median, the back-transformed mean prediction.

Aggregate has a median of 0.76 % at the set-point, and its single-run limits over the NOR run from 0.49 % to 1.40 %. The least favourable combination is protein load 32 g/L resin, load/wash conductivity 6 mS/cm, elution buffer pH 6.15 and elution stop collect 1.3 AU. Host cell protein has a median of 159 ng/mg at the set-point and single-run limits of 86.5 to 289 ng/mg, the least favourable combination being protein load 32 g/L resin, load/wash conductivity 4 mS/cm and elution buffer pH 6.15. Elution stop collect is not in the host cell protein model, so that limit holds at any stop collect setting. No predicted value over the NOR comes near its criterion. At the least favourable combinations the median lies 9 SDs of the ELISA below the host cell protein criterion and 92 SDs of the SEC method below the aggregate criterion, so the plan's rule for a value within assay variability of its criterion does not come into play. Fragments are judged on their measured results, all below their criterion (Section 5.3.5).

7.3 Proven acceptable ranges per parameter

The set-point analysis gives the range of one parameter with the rest of the step at target. The NOR-propagated analysis gives the range over which the 97.5th percentile of simulated single batches meets the criterion while the other factors vary within their NORs, and it is the range that supports a claim (Section 3.5). Table 7.3 gives both for aggregate and host cell protein. Every range is the range studied, so each proven acceptable range reaches the edge of the characterized range and no claim is made beyond it (PCMP-001 §7.6). Fragments have no adequate model, and no proven acceptable range is derived for them.

Table 7.3: Proven acceptable ranges of the multivariate parameters for aggregate and host cell protein.

Attribute	Criterion	Parameter	Unit	PAR at set-point	PAR, NOR-propagated
Aggregates, HMW (%)	≤ 4.90	Protein load	g/L resin	10–40	10–40
		Load/Wash conductivity	mS/cm	3–7	3–7
		Elution buffer pH	pH	5.8–6.2	5.8–6.2
		Elution stop collect	AU (OD280, descending)	0.5–1.5	0.5–1.5
Host cell protein (ng/mg)	≤ 418	Protein load	g/L resin	10–40	10–40
		Load/Wash conductivity	mS/cm	3–7	3–7
		Elution buffer pH	pH	5.8–6.2	5.8–6.2
		Elution stop collect	AU (OD280, descending)	0.5–1.5	0.5–1.5

Each range is the range studied of the parameter. Criterion in the unit of the attribute.

Since each range is the range studied, each NOR edge lies inside its proven acceptable range by the distance to the end of the range studied, 8 g/L resin for protein load, 1 mS/cm for load/wash conductivity, 0.05 pH units for elution buffer pH and 0.2 AU

for elution stop collect. Figure 7.1 plots aggregate against protein load, which has the largest main effect on it. For host cell protein the main effects of load and conductivity are of the same size within their standard errors, so Figure 7.2 and Figure 7.3 plot both.

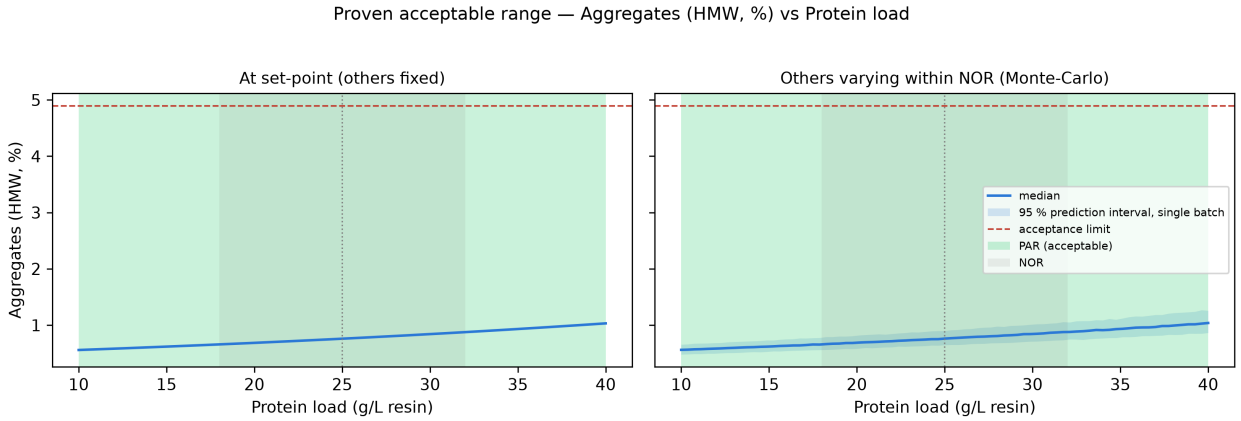


Figure 7.1: Proven acceptable range of protein load for aggregate, at manufacturing scale. Left, the set-point analysis. Right, the NOR-propagated analysis with the 95 % interval of a single batch. The acceptable region is shaded green and the NOR grey.

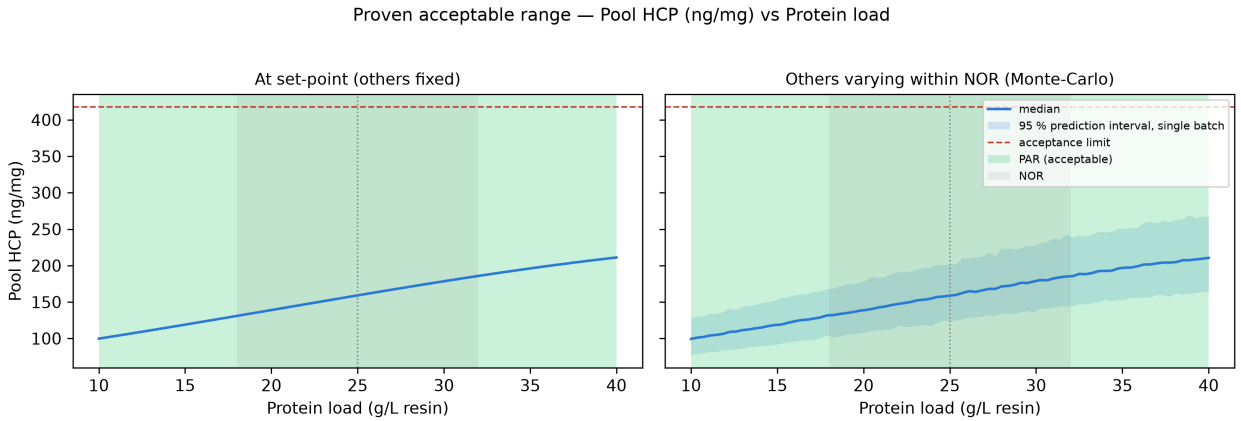


Figure 7.2: Proven acceptable range of protein load for host cell protein, at manufacturing scale. Left, the set-point analysis. Right, the NOR-propagated analysis with the 95 % interval of a single batch.

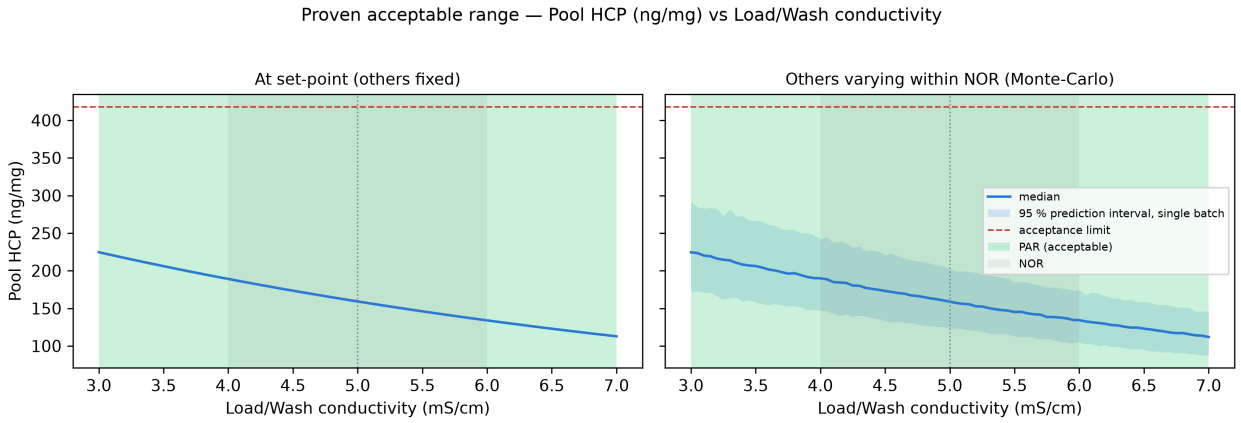


Figure 7.3: Proven acceptable range of load/wash conductivity for host cell protein, at manufacturing scale. Left, the set-point analysis. Right, the NOR-propagated analysis with the 95 % interval of a single batch.

A proven acceptable range is a statement about one parameter while the others are held at defined settings. The design space is a statement about the four together, and a combination of settings inside their separate ranges could fall outside it where an interaction is active. At this step, at the criteria in force, both cover the whole range studied. For elution flow rate, the range of 100–300 cm/hr is supported by its own data for aggregate and host cell protein (Section 5.5).

7.4 Verification of the platform worst cases

The criteria upstream of this step assume that cation exchange clears each attribute at least as well as its platform worst case. Table 7.4 sets each platform worst case against this campaign's results.

Table 7.4: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign			Outcome
				At the edges	Within the NOR	95 % bound	
Cation exchange	Host cell protein	Clearance factor, lowest	40.0	27.2	40.2	32.9	Confirmed
	Aggregates, HMW		2.00	1.00	1.58	1.40	Not confirmed, tightened[a)]
	Fragments, LMW		1.30	1.28	1.45	1.27	Confirmed
	Leached Protein A		3.16	not a response[b)]			

[a)] The criteria upstream of this step were recalculated with the one-sided 95 % prediction bound of a single batch at the value predicted within the NOR.
[b)] Not a response of the response-surface study at this step, so the platform value stands unverified.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

Aggregate. Within its NOR the step cleared aggregate less than the platform worst case of 2.00. At manufacturing scale the lowest clearance factor predicted was 1.58, at the combination of protein load 32 g/L resin, load/wash conductivity 6 mS/cm, elution buffer pH 6.15 and elution stop collect 1.3 AU. By rule 1 of PCMP-001 §7.5 the one-sided 95 % prediction bound of a single batch there, 1.40, replaced the platform value, and the aggregate criteria upstream became stricter. The criterion of the low-pH pool moved from 10.0 % to 6.84 %. The criterion of the bioreactor harvest moved from 9.30 % to 6.30 %, a value that also includes a widening from the low-pH step (PCR-006). The bound comes from the 28 runs of the response-surface design, on a model that meets all four adequacy criteria. At the edges of the range studied the step cleared less still, 1.00 at the lowest, outside the NOR. With each flagged run deleted the lowest predicted factor stays below the platform value and the bound lies between 1.39 and 1.43 (Table 5.8). The plan's full quadratic would have given a bound of 1.34.

That tightening is argued by its risk, in the four parts of PCMP-001 §7.5.

1. Who bears the risk. A stricter criterion lowers the aggregate that the low-pH pool and the harvest may carry. It lowers the patient's risk and raises the producer's, because a pool between the new and the a priori criterion now fails.
2. Detection. No step after cation exchange removes aggregate, so the aggregate of this pool reaches the drug substance with the small rise across UF/DF. There it is a release test by SEC. An excursion of a pool upstream beyond its tightened criterion would therefore be detected at release whenever it carried the drug substance above 5.00 %.
3. What the later steps must clear. Cation exchange must clear 1.40-fold from the criterion of the low-pH pool to the criterion of its own pool, 4.90 %, which is its own bound within the NOR. UF/DF may then add up to 0.104 percentage points, which brings the pool to the drug substance criterion. No later step is credited with any clearance of aggregate.
4. The drug substance. No simulated batch exceeded the aggregate criterion of the drug substance, so the probability is below 0.15 %, and the capability index is 15.7 (Section 8).

Two studies of the low-pH viral inactivation report ran this step at its set-point on material from the low-pH step, after the criteria were fixed. They tested whether cation exchange clears aggregate and fragments by the factors the criteria rely on, 1.40 and 1.30. In S-CEX-VER (discrepancy PD-006-02) and S-VI-WC (PD-006-03) every run cleared aggregate by 2.21 to 2.67-fold, and the lower confidence bound of each study met the factor required. For fragments, S-CEX-VER met its factor. In S-VI-WC every run cleared fragments by more than 1.30-fold, but the lower bound of the three runs did not, so the rule set before those runs was not met, and PCR-006 accepts the result on its risk. Both studies, their results and their decisions belong to PCR-006.

Host cell protein. At manufacturing scale the lowest clearance factor predicted within the NOR, 40.2, is not below the platform worst case of 40.0, so the assumption is confirmed. The bound of a single batch, 32.9, is not better than the platform value, so the platform value stands. At the edges of the range studied the lowest factor measured was 27.2, outside the NOR. The same outcome follows with each flagged

run deleted. It does not follow under every model within 2 AICc units of the selected one. Under 3 of the 5 other models the lowest factor falls below the platform value (Section 5.3.4).

The verification converts the scale-down prediction to manufacturing scale with the offset at its estimate (PCMP-001 §7.1 rule 2). Its outcome for host cell protein depends on that estimate. With the offset at the unfavourable end of its interval the lowest factor would be 28.1 and the bound 23.0, and rule 1 of PCMP-001 §7.5 would then have tightened the host cell protein criterion of the Protein A pool. With the offset at the favourable end the lowest factor would be 57.6 and the bound 47.1, better than the platform value. For aggregate the outcome is the same at either end of its interval, with lowest factors of 1.40 and 1.78.

Fragments. The lowest clearance factor that the fragment model predicts within the NOR, 1.45, is not below the platform worst case of 1.30. By the rule of PCP-007 §7.6 the assumption is therefore confirmed, as Table 7.4 shows. The bound of a single batch, 1.27, is not better than the platform value, so the platform value stands. The fragment model behind this confirmation does not meet the limit on overfitting of PCMP-001 §7.2 (Section 5.3.5), and fragments have no measured offset, so the factor is the one the scale-down model gives. At the edges of the range studied the lowest clearance factor measured was 1.28, just below the platform value. The fragment criteria of the Protein A pool (2.45 %) and the low-pH pool (2.60 %) rely on this value.

Leached Protein A. Leached Protein A is not a response of the designs, so its platform value of 3.16 stands without verification by a model. Every response-surface run cleared more than that factor (Section 5.4).

8 Process capability and robustness

Capability is estimated for the drug substance from the integrated model of the train (PCMP-001 §6.5). The capability of the step itself, which PCP-007 §5.5 asked for, was not estimated (Section 3.5). Table 8.1 gives the capability index of each drug substance attribute that this step clears, from 2,000 simulated batches with every parameter drawn within its NOR.

Table 8.1: Capability of the drug substance attributes that cation exchange clears.

Attribute	Criticality	Drug substance criterion	Mean ± SD	Cpk	Simulated batches outside
Aggregates, HMW (%)	H	≤ 5.00	0.785 ± 0.090	15.66	0
Host cell protein (ng/mg)	M-H	≤ 100	15.2 ± 4.6	6.14	0
Fragments, LMW (%)	L	≤ 2.00	0.643 ± 0.043	10.62	0
Residual DNA (ng/dose)	VL	≤ 0.001	0.000170 ± 0.000038	7.30	0
Leached Protein A (ppm)	L-M	≤ 5.00	0.00298 ± 0.00060	2,785	0

One-sided index against the upper criterion. Mean and SD of 2,000 simulated batches of the integrated model, in the unit of the criterion. Residual DNA in ng per dose.

Every index exceeds the target of 1.33, and none approaches the minimum of 1.0. Host cell protein has the lowest index, 6.14, with a mean of 15.2 ng/mg against 100 ng/mg. No simulated batch fell outside any of the five criteria. With 2,000 batches that bounds the probability of an out of specification result for each attribute below 0.15 % at one-sided 95 % confidence. The bound covers the sampling of the simulation and not the uncertainty of the integrated model itself.

The index of leached Protein A, 2,785, is not meaningful, because the simulated mean of 0.0030 ppm lies far below the criterion of 5.00 ppm. Leached Protein A is judged instead by the criterion of the Protein A pool and the clearance of the steps after it. Residual DNA is judged per dose. The dose is A-Mab’s 10 mg/kg, 700 mg for a reference adult of 70 kg, and the body weight is an assumption of this campaign. The simulated drug substance carries 0.00017 ng per dose on average against 0.001 ng. That result depends on inputs from outside this step’s data, the DNA clearance that the process model credits to this step (1.5 log10) and to anion exchange (5 log10). The pool of this step alone carried up to 43.6 pg/mg in the response-surface runs, 30.5 ng per dose, so the drug substance criterion is met through anion exchange.

The derivation of the criteria credits no later step with removing aggregate, so the aggregate of the drug substance is the aggregate of this pool plus the small rise

across UF/DF. It shares the clearance of host cell protein, residual DNA and leached Protein A with anion exchange. The robustness of the step within its NOR is shown by the margins of Section 7.2, and its least margin lies at corners outside the NOR (Section 6).

9 Parameter classification

Parameters were classified under SOP-4001 by the decision logic of the A-Mab case study (CMC Biotech Working Group 2009), as PCMP-001 §7.7 sets it out. A critical process parameter "has an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation". A well-controlled critical process parameter "has the same kind of effect on a CQA but a low risk of falling outside its acceptable range, because it is readily controlled and its PAR is wide against its NOR". A key process parameter affects process performance and has no effect on a CQA within its range. A general process parameter "has no relevant effect on a CQA or on process performance within its range". PCMP-001 §7.7 adds that "a significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical". An attribute is called affected by a parameter when a term that contains the parameter is significant at $\alpha = 0.05$ (PCMP-001 §6.2). No numerical threshold is set for practical relevance or for a PAR that is wide against its NOR. Each class below is therefore stated with the size of the effect across the proven acceptable range and the margin of that range to the NOR.

- Protein load was classified as a WC-CPP. It affects aggregate, through its main effect and its interactions with pH and stop collect, and host cell protein, through its main effect. With the other factors at their set-points, aggregate rises from 0.56 % to 1.03 % and host cell protein from 99.7 to 211 ng/mg across the range of 10–40 g/L resin. Its proven acceptable range is the range studied, 8 g/L resin beyond each end of its NOR, and the load is set by the mass of antibody applied per litre of resin.
- Load/wash conductivity was classified as a WC-CPP. It affects host cell protein, through its main effect and its interaction with pH, and has no demonstrated effect on aggregate. Host cell protein falls from 225 to 113 ng/mg across 3–7 mS/cm. Its proven acceptable range is the range studied, 1 mS/cm beyond each end of its NOR.
- Elution buffer pH was classified as a WC-CPP. It affects aggregate, through its main effect and its interactions with load and stop collect, and host cell protein through its interaction with conductivity. Aggregate rises from 0.64 % to 0.90 % across pH 5.8–6.2. Its proven acceptable range is the range studied, 0.05 pH units beyond each end of its NOR, the narrowest margin of the four.
- Elution stop collect was classified as a WC-CPP. It affects aggregate, through its main effect, its curvature and its interactions with load and pH, in the direction opposite to the expectation (Section 5.6). Aggregate is 0.72 %, 0.76 % and 0.99 % at settings of 0.5 AU, 1 AU and 1.5 AU. Its proven acceptable range is the range studied, 0.2 AU beyond each end of its NOR.

- Elution flow rate was classified as a GPP. The class rests on a judgement of practical relevance, for which no plan sets a threshold. Its range studied of 100–300 cm/hr extends 50 cm/hr beyond each end of its NOR. No difference was detected in aggregate or host cell protein, and every result met its criterion (Section 5.5). Residual DNA clearance was 0.32 log10 lower at 300 cm/hr than at 100 cm/hr ($p = 0.044$), so under PCMP-001 §6.2 residual DNA is affected by flow rate. The difference is a factor of 2.1 in the residual DNA of the pool across the range studied, and about 1.45 across the NOR if the effect is linear in flow rate. The drug substance carries 0.00017 ng per dose on average against a criterion of 0.001 ng, with a capability index of 7.30 (Section 8). If the whole factor reached the drug substance, its mean would be 0.00036 ng per dose and its capability index 2.67, still above the target of 1.33. This report therefore judges the effect too small to move residual DNA appreciably in the drug substance, in the sense of PCMP-001 §7.7. For step yield the study detected no difference, but the interval of the difference, −5.19 to 2.00 points, includes an effect as large as that of protein load (−5.18 points). The study therefore does not show that flow rate leaves yield unchanged. For yield the class rests on the absence of a detected difference and on the expectation of PCP-007 §4.1 that flow rate, within its range, changes the sharpness of the elution peak only a little. Fragments were not measured in the flow-rate study, so for fragments the class rests on the mechanism alone. Flow rate changes the residence time and the mass transfer in the resin, and it changes neither the charge of the antibody nor the ionic strength that holds it.

Of the 5 parameters, 4 are critical process parameters, all of them well controlled, and one is a general process parameter. Each WC-CPP has a proven acceptable range equal to its range studied, and the conclusions over the NOR hold with margin for each attribute with a criterion.

10 Contribution to the control strategy

The step contributes four elements to the control strategy of the drug substance. The results of this report roll up, with those of the other steps, into the master report PCMR-001.

1. Parameter ranges. The four WC-CPPs are controlled within their NORs, which lie inside proven acceptable ranges equal to the ranges studied and inside a design space that covers the whole region studied at the criteria in force (Section 6 and Section 7.3). Elution flow rate is controlled within its NOR as a GPP.
2. Criteria for the steps upstream. The verification of this report tightened the aggregate criteria of the low-pH pool and the bioreactor harvest, which PCR-006 and PCR-003 judge. The host cell protein criterion of the Protein A pool, 16,719 ng/mg, assumes the platform clearance of this step, which this report confirmed with the offset at its estimate (Section 7.4).
3. The criteria this step hands on. The pool must meet 418 ng/mg of host cell protein for anion exchange to bring it to the drug substance criterion. It must meet 4.90 % of aggregate so that the rise across UF/DF keeps the drug substance within its criterion. It meets both across the NOR, with highest single-run limits of 289 ng/mg and 1.40 %.
4. Release testing. The drug substance criteria of Table 2.1 apply at release. For aggregate and fragments the register names the method, SEC and nrCE-SDS.

This report proposes no in-process control or monitoring of the pool beyond these four elements.

The conclusions of this report hold for resin within the cycle count qualified under SOP-2008. Cation exchange claims no viral clearance. Low-pH viral inactivation inactivates enveloped virus (PCR-006). Anion exchange clears host cell protein, residual DNA, leached Protein A and virus after this step (PCR-008). Virus filtration removes virus (PCR-009).

11 Discussion

A higher protein load and a higher elution buffer pH raise aggregate in the pool. More load broadens the elution peaks, so that aggregate overlaps more of the monomer. A higher pH lowers the positive charge that holds both species, so that they come off closer together. A higher protein load also raises host cell protein, because more of it is held with the bound antibody. A higher load/wash conductivity lowers host cell protein, because a higher ionic strength strips more of the weakly bound species in the wash, and elution buffer pH changes the size of that effect. A higher stop collect setting raises aggregate, against the expected mechanism. The selected yield model has no stop collect term, where a later stop was expected to recover more monomer. Neither result is explained.

Confidence in the scale-down model rests on the offsets of Table 3.1. For aggregate the offset is close to 1 and well determined. For host cell protein the ratio is known only to within its interval of 0.63 to 1.28. The model is qualified for host cell protein at the criterion in force. It would not be qualified at the a priori criterion, under 3 of the other models within 2 AICc units of the selected one, or with the run-order drift added at the unfavourable end of the offset interval (Section 3.1, Section 5.3.4 and Section 5.3.2).

Screening run 6 was made with the expired buffer lot. With it deleted, the screening models find the same significant terms. Response-surface run 14 had the temperature excursion. With it deleted, the judgement over the NOR, the design space and the outcome of the verification are unchanged (Section 13). Deleting run 14 does change two selected models. The aggregate model loses its conductivity term, and the yield model replaces its protein load × load/wash conductivity interaction with a stop collect term.

The discrepancies from the plan fall into three groups. The analysis departs from the plan in the model selection, in the use of a host cell protein model that does not meet the residual criterion, and in the approaches taken where the plan sets no threshold. The selection changed the terms of the models but no conclusion over the NOR. The run-order drift of the host cell protein residuals, a factor of 1.13 over the study, would raise the highest single-run limit over the NOR to 326 ng/mg, below the criterion of 418 ng/mg. The execution departs from the plan in the flow-rate study without its set-point run, the residual DNA offset recorded without investigation, and the fractionated runs of S-CEX-FRC. The record lacks the load results, the residual DNA and leached Protein A results of the screening runs, the fractions and the mass balance, the packing tests of each campaign and the feed lots. The missing load results matter most, because every clearance factor of aggregate and host cell protein rests on the process model at set-point instead of a measured load. Apart from these groups,

discrepancy PD-007-12 corrects the name of the load/wash buffer of screening run 6 and changes no result.

Five limitations remain.

1. Scale. The scale-down model does not reproduce the hold times, pool volumes or mixing of the commercial suite, and no run at manufacturing scale tests the edges of the design space. The process performance qualification (PPQ) batches of Stage 2, which have not yet been run, test the step at its set-points.
2. The corner of highest load and lowest conductivity. One screening result there exceeded the host cell protein criterion, and the edge of the design space at that corner depends on the choice among the models within 2 AICc units of the selected one and on the run-order drift. The corner lies outside the NOR of every parameter.
3. The size of the flow-rate study. With two runs at each end and no set-point run, the study cannot exclude an effect on yield as large as that of load, and its significant difference in DNA clearance rests on four runs.
4. Fragments. Fragments have no adequate model and no offset. Their judgement rests on measured results in the scale-down model and on the assumption that they separate alike at both scales, and the confirmation of their platform worst case rests on the inadequate model.
5. Resin life cycle. The conclusions hold within the qualified life of the resin, and performance at the end of that life is established by the life-cycle study under SOP-2008, not here.

12 Conclusions

1. Objective 1 is met. The relationships between the four multivariate parameters and the outputs of the step were quantified over ranges wider than the NORs. Protein load, elution buffer pH and elution stop collect affect aggregate, and protein load and load/wash conductivity affect host cell protein, with pH acting through conductivity. The stop collect acts on aggregate in the direction opposite to the expectation.
2. Objective 2 is met for aggregate and host cell protein and not met for fragments. The design space is drawn on the single-run limit, at manufacturing scale, for aggregate and host cell protein. At the criteria in force it covers the whole region studied, and at the a priori host cell protein criterion it would cover 99.85 %. Host cell protein comes nearest to its criterion, at a corner outside the NOR. Fragments have no adequate model, so no model predicts where they meet their criterion. Their measured results, all below the criterion, constrain the region instead.
3. Objective 3 is met for aggregate and host cell protein and not met for fragments. Every output meets its intermediate acceptance criterion across the NOR. The highest single-run limits are 1.40 % for aggregate and 289 ng/mg for host cell protein, and fragments were 0.76 % at most. Every proven acceptable range of aggregate and host cell protein is the range studied. Fragments have no adequate model and therefore no proven acceptable range.
4. Objective 4 is met. The four multivariate parameters were classified as WC-CPPs and elution flow rate as a GPP, the latter by a judgement of the practical relevance of its effect on residual DNA.
5. Objective 5 is met. The step contributes its parameter ranges, the criteria it sets for the steps upstream and the criteria it hands on to the control strategy. No in-process control or monitoring of the pool is proposed beyond these. The capability indices of the drug substance attributes it clears exceed 1.33, the lowest 6.14 for host cell protein.
6. Objective 6 is met. The platform worst case for aggregate was not confirmed, and its bound of 1.40 tightened the aggregate criteria upstream. The worst case for host cell protein was confirmed with the offset at its estimate and would not be at the unfavourable end of its interval. The worst case for fragments was confirmed by the rule of PCP-007 §7.6, because the lowest clearance factor predicted within the NOR is not below it, but on a model that does not meet PCMP-001 §7.2. The worst case for leached Protein A stands without verification by a model.
7. The runs of both study events were retained. Without screening run 6, made with the expired buffer lot, the screening models find the same significant terms. Without response-surface run 14, made during the temperature excursion, the judgement over the NOR, the design space and the outcome of the verification

are unchanged, while the selected models of aggregate and yield change terms. The register records 12 discrepancies from the plan.

13 Study events and discrepancies from the plan

A process characterization study is not manufacture under good manufacturing practice, so it raises no deviation. The Version 1.0 plans called a departure from the plan a deviation. This report records two kinds of entry instead. Study events are the events of execution that Version 1.0 recorded, kept under the DEV numbers Version 1.0 gave them. Discrepancies from the plan are departures from the approved plan PCP-007, whether work the plan did not foresee or a requirement of the plan that was not met.

13.1 Study events

The studies of this step recorded 2 study events, and the runs concerned were retained in the analysis.

Table 13.1: Study events of the cation exchange studies.

Event	Summary	Detected during	Outcome
DEV-007-01	Expired acetate load/wash buffer lot used in screening run 6	post-execution documentation review	run retained in the analysis
DEV-007-02	Column inlet temperature excursion during RSM run 14	equipment history review	run retained in the analysis

13.1.1 DEV-007-01 Expired buffer lot in a screening run

The load/wash buffer used in screening run 6 came from lot LOT-BUF-2287, which had expired on 2026-02-11, before the screening study started on 2026-02-16. The lot is an acetate load/wash buffer, and Version 1.0 of this report named it a Tris buffer in error (discrepancy PD-007-12). The expiry was found in the documentation review after the study. The record gives the cause as an expiry date not checked at preparation. A lot that is within its expiry date when it is prepared can still be expired when it is used, so the check that would have found this lot is a check at the time of use. Under SOP-2103 (Buffer Preparation, Release and Expiry Control), which authorises a re-test of the retained sample for an expired lot, the retained sample was

re-tested and gave pH 5.02 and a conductivity of 5.06 mS/cm. That pH lies in the buffering range of acetate. The record gives no acceptance range for these values, so this report does not judge the lot from them.

The impact was judged from the data. Run 6 was made at protein load 10 g/L resin, load/wash conductivity 7 mS/cm, elution buffer pH 5.8 and elution stop collect 1.5 AU. Its studentized residuals on the screening models were 1.24 for aggregate, 0.18 for host cell protein and −0.48 for step yield. With the run excluded, the screening models find the same significant terms for all three outputs. The screening design identifies effects and sets no range, so the run was retained.

13.1.2 DEV-007-02 Temperature excursion during a response-surface run

During response-surface run 14 the column inlet temperature was above its set-point for 38 minutes and reached 26.4 °C. The record gives neither the temperature set-point nor a permitted range. The excursion was found in the equipment history review of the chamber EQ-CHR-118, and its cause was a drift of the set-point of the chamber controller. Temperature is not a factor of the designs. Aggregate in the pool of the run was verified by 3 determinations by SEC under the method validation AMV-3221. At its precision of 3.8 % RSD, the mean of 3 determinations has a 95 % confidence interval of ± 9.44 % of its value. The record gives neither the result of these determinations nor a criterion they were judged against, so the verification decides nothing in this report. The impact of the run was judged from its residuals instead.

Run 14 was made at protein load 40 g/L resin, load/wash conductivity 7 mS/cm, elution buffer pH 5.8 and elution stop collect 1.5 AU. Its studentized residuals on the selected models were 0.23 for aggregate, −0.92 for host cell protein, 0.34 for fragments and 1.11 for step yield. With the run excluded and the selection repeated, the host cell protein model keeps the same terms and the aggregate model loses its conductivity main effect, which was not significant. Both outputs still meet their criteria over the NOR, with highest single-run limits of 1.39 % and 288 ng/mg, and the whole region studied still meets them. The lowest predicted clearance factor of aggregate stays below the platform value. The step yield model replaces its protein load × load/wash conductivity interaction with a stop collect term (Section 5.3.6), and yield has no criterion. The run was retained.

13.2 Discrepancies from the plan

Table 13.2 lists the discrepancies from PCP-007.

Table 13.2: Discrepancies from the plan PCP-007.

Discrepancy	Subject	Planned	Done	Reason
PD-007-01	Model selection for the response-surface outputs	PCP-007: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report

Discrepancy	Subject	Planned	Done	Reason
PD-007-02	Conclusions resting on models that do not meet the criterion on residuals and influential runs	PCP-007: response-surface models meeting every model-adequacy criterion of PCMP-001 §7.2, used for the ranges and the design space, and each run with a Cook’s distance above 4/n examined by deleting it and repeating the judgement	the models of the outputs that do not meet the criterion on residual patterns and influential runs (a residual trend at the level of the residual tests, or a run with a Cook’s distance above the median of F(p, n − p)) used for the ranges and the design space, with the size of each trend stated against the criterion of its output and each run above the limit deleted in turn, the model selection repeated and the output judged over the NOR again	selected models of this step do not meet the criterion of PCMP-001 §7.2 on residual patterns and influential runs; whether a conclusion depends on the trend or on the run is shown by the size of the trend and by the deletion
PD-007-03	Flow-rate study without the run at the set-point	PCP-007: the flow-rate study run at the ends of the range and at the set-point	each end of the range run in duplicate, with no run at the set-point; the qualification runs at set-point serve as the reference	the set-point reference was taken from the qualification runs

Discrepancy	Subject	Planned	Done	Reason
PD-007-04	Residual DNA offset of the scale-down model recorded, not investigated	PCP-007: Version 1.0: no statistically significant difference on any compared attribute, residual DNA among them, and the cause of a difference established and the model modified before charac-terization proceeds	residual DNA differed between the scales at that level; it was not investigated before the characteriza-tion runs, and its offset is recorded only, because residual DNA has no criterion at this step (PCMP-001 Version 2.0 §7.1)	a scale-down model may differ in mean from manufacturing scale where its offset is known precisely enough, the rule of PCMP-001 Version 2.0 §7.1, so the Version 1.0 test was stricter than the use of the model needs
PD-007-05	Load aggregate and host cell protein not recorded	PCP-007: aggregate and host cell protein measured in the load of every run, so that each output is read as a fold or an increase across the step (PCP-007 §5.4, §12)	no load result for aggregate or host cell protein is in the record; the folds are taken against the load levels of the process model; Version 1.0 stated that the load was assayed, with no result recorded	the record gives no reason
PD-007-06	Residual DNA and leached Protein A not measured on the screening runs	PCP-007: residual DNA and leached Protein A measured on every run of the screening and response-surface designs (PCP-007 §5.4)	measured on the response-surface runs only	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-007-07	Flow-through, wash and strip fractions and the mass balance not recorded	PCP-007: the flow-through, wash and strip fractions sampled and a mass balance made for every run, with the strip analysed when the mass balance falls short (PCP-007 §5.4)	no fraction result and no mass balance is in the record, so the strip test could not be triggered; Version 1.0 stated that these fractions were assayed, with no result recorded	the record gives no reason
PD-007-08	Packing tests not recorded for each campaign	PCP-007: plate count and asymmetry tested and the column released against SOP-2008 before each campaign (PCP-007 §5.1)	plate count and asymmetry recorded for the three qualification runs only, with no acceptance limits; no packing result for the screening, response-surface and flow-rate campaigns; Version 1.0 stated that the column was released, with no result recorded	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-007-09	Fractionated runs not in the plan	PCP-007: the studies of PCP-007 Table 11.1	study S-CEX-FRC, runs with the eluate collected in fractions, was made and is reported; it is in neither version of the plan	the fractions were collected to read the elution profile against stop collect
PD-007-10	Approaches where the plan sets no threshold, and step-level capability	PCP-007: the decisions of PCP-007 §7 and §8 made on the evidence named there, and the capability of the step estimated from the fitted models over 2,000 draws within the NOR, as a Cpk for each attribute of Table 7.1 against its intermediate criterion (PCP-007 §5.5)	the report states its own approach for the decisions the plan leaves without a threshold, and three of them could not be made as planned because the load results, the fractions and the packing records are missing (PD-007-05, PD-007-07, PD-007-08); capability is given for the drug substance from the integrated model, not for the step	the plan names the evidence for these decisions but no threshold; the capability of the drug substance is the estimate PCMP-001 Version 2.0 §7.9 uses

Discrepancy	Subject	Planned	Done	Reason
PD-007-11	Feed lots not recorded	PCP-007: the feed lot of every run recorded and the lots compared where more than one is used (PCP-007 §6.2)	the record does not name the feed lot of each run	the record gives no reason
PD-007-12	Load/wash buffer named Tris in Version 1.0 of the report	PCP-007: the materials of each run, the load/wash buffer among them, reported as the buffer record names them	the acetate load/wash buffer was used (lot LOT-BUF-2287 in screening run 6, study event DEV-007-01); Version 1.0 of this report named it a Tris buffer in error, and its re-test pH lies in the buffering range of acetate	a transcription error in Version 1.0 of the report, found in the Version 2.0 review

13.2.1 PD-007-01 Model selection

PCP-007 prescribed the full quadratic model of each response-surface output. Each output was analysed instead by the model selection of INV-002 corrective action CA3 (Section 3.5), because the full quadratic carries terms the data do not support. The selection changed the terms of every model and put aggregate and host cell protein on the log scale. It changed no conclusion over the NOR. With the plan's full quadratic, aggregate and host cell protein also meet their criteria over the NOR, and fragments fail the limit on overfitting under either analysis. Through the anion exchange model, the selection also widened the host cell protein criterion of this step from 400 to 418 ng/mg. That widening decides the qualification of the scale-down model for host cell protein and whether the design space covers the whole region studied, and no conclusion over the NOR (Section 7.1). It also moved the bound that tightened the aggregate criteria upstream, from 1.34 under the full quadratic to 1.40.

13.2.2 PD-007-02 Conclusions on a model that does not meet the residual criterion

PCP-007 required models that meet every adequacy criterion of PCMP-001 §7.2. The host cell protein model meets the first three and fails the fourth, because its residuals fall with the order of execution ($p = 0.043$). It was used for the ranges and the design space. The trend amounts to a factor of 1.13 over the study, and added in full to the highest single-run limit over the NOR it gives 326 ng/mg against the criterion of 418 ng/mg (Section 5.3.2). No run of any model has a Cook's distance above the median of $F(p, n - p)$, so no run fails the criterion. The flagged runs were each deleted and the selection repeated, which the plan provides, and no judgement over the NOR changed (Section 5.3.3). Outside the NOR the drift would move the edge of the design space and the qualification for host cell protein (Section 5.3.2).

13.2.3 PD-007-03 Flow-rate study without the set-point run

PCP-007 §6.4 required the flow-rate study at the ends of the range and at the set-point. Each end was run in duplicate, and no run was made at the set-point. Table 5.10 shows the qualification runs at the set-point beside the flow-rate runs. They were made in a different study, before the characterization runs, so a difference between them and the flow-rate runs mixes any effect of flow rate with the variation between the two studies. The comparison of the two ends does not depend on them.

13.2.4 PD-007-04 Residual DNA offset recorded, not investigated

Version 1.0 of PCP-007 accepted the scale-down model only where no compared attribute differed significantly between the scales, and required the cause of a difference to be established before characterization went on. Residual DNA differed at that level ($p = 0.016$), and the difference was not investigated before the characterization runs. Under PCMP-001 Version 2.0 §7.1 a scale-down model may differ in mean from manufacturing scale where the offset is known precisely enough, so the Version 1.0 test was stricter than the use of the model needs (Section 3.1).

13.2.5 PD-007-05 Load aggregate and host cell protein not recorded

PCP-007 §5.4 and §12 required aggregate and host cell protein to be measured in the load of every run, so that each output is read as a fold across the step and a clearance can be told apart from a low starting level. No load result for either attribute is in the record. Version 1.0 of this report stated that the load was assayed, with no result recorded. The clearance factors of this report are taken against the process model at set-point, which brings 1.91 % aggregate and 9,125 ng/mg host cell protein to the step. The pool results and the judgements against the criteria do not depend on the load. The clearance factors and the verification of the platform worst cases do, and a load that differed from the process model between runs would appear in them as scatter or as a trend.

13.2.6 PD-007-06 Residual DNA and leached Protein A not measured on the screening runs

PCP-007 §5.4 required residual DNA and leached Protein A on every run of both designs. They were measured on the response-surface runs only. The analysis of both attributes rests on the 28 response-surface runs (Section 5.4), and the flow-rate comparison of residual DNA has no screening effect to set against it (Section 5.5).

13.2.7 PD-007-07 Flow-through, wash and strip fractions and the mass balance not recorded

PCP-007 §5.4 required the flow-through, wash and strip fractions to be sampled and a mass balance made for every run, with the strip analysed when the mass balance falls short. No fraction result and no mass balance is in the record, so the strip test could not be triggered. Version 1.0 of this report stated that the flow-through and the wash were assayed, with no result recorded. Where host cell protein is removed within the step, in the load, the wash or the strip, therefore cannot be shown, and the mechanism of Section 2.1 is not tested on that point.

13.2.8 PD-007-08 Packing tests not recorded for each campaign

PCP-007 §5.1 required plate count and asymmetry to be tested, and the column released against SOP-2008, before each campaign. Plate count and asymmetry are recorded for the 3 scale-down runs of the qualification only, with no acceptance limits. No packing result is recorded for the screening, response-surface and flow-rate campaigns. Version 1.0 of this report stated that the column was released, with no result recorded. A change in packing between or within campaigns therefore cannot be excluded from the record.

13.2.9 PD-007-09 Fractionated runs not in the plan

The study S-CEX-FRC was made and is reported, and it is in neither version of the plan. Its fractionated run collected the eluate in fractions to read the elution profile against the stop collect, and its samples were analysed by SEC and by nrCE-SDS (Section 3.3). Its results support the method decision for fragments and are not used for any range or class of this step.

13.2.10 PD-007-10 Approaches where the plan sets no threshold, and step-level capability

PCP-007 §7 and §8 name the evidence for seven decisions but no threshold. This report states and applies its own approach for each (Section 3.5.1). Three of the decisions could not be made as planned, because the packing records, the fractions

and the load results are missing (PD-007-08, PD-007-07 and PD-007-05). PCP-007 §5.5 also asked for the capability of the step, from the fitted models over draws within the NOR, as a Cpk for each attribute against its intermediate criterion. Capability is given for the drug substance from the integrated model instead (Section 8), which is the estimate PCMP-001 Version 2.0 §7.9 uses.

13.2.11 PD-007-11 Feed lots not recorded

PCP-007 §6.2 required the feed lot of every run to be recorded and the lots compared where more than one is used. The record does not name the feed lot of any run. A lot effect therefore cannot be separated from the run-order drift of host cell protein or from the scatter between runs.

13.2.12 PD-007-12 Load/wash buffer named Tris in Version 1.0

Version 1.0 of this report named the load/wash buffer of lot LOT-BUF-2287, used in screening run 6, a Tris buffer (Section 13.1.1). The lot is an acetate load/wash buffer. The name in Version 1.0 was a transcription error, found in the Version 2.0 review. The re-test pH of the lot, 5.02, lies in the buffering range of acetate. The correction changes no result of this report.

Appendix A. Screening design matrix and responses

Table A.1 gives the settings and results of every screening run, in natural units and in run order of the design.

Table A.1: Cation exchange screening design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW
		A g/L resin	B mS/cm	C pH	D AU	HMW %	HCP ng/mg	Yield %	SEC %
1		10	3	5.8	0.5	0.50	136	93.6	0.59
2		10	3	5.8	1.5	0.60	114	96.2	0.71
3		10	3	6.2	0.5	0.55	135	94.0	0.54
4		10	3	6.2	1.5	0.82	138	97.3	0.66
5		10	7	5.8	0.5	0.45	66.2	93.4	0.60
6[a)]		10	7	5.8	1.5	0.56	61.3	95.1	0.75
7		10	7	6.2	0.5	0.58	48.8	94.6	0.56
8		10	7	6.2	1.5	0.75	67.6	93.4	0.67
9	factorial	40	3	5.8	0.5	0.79	428	88.3	0.58
10		40	3	5.8	1.5	1.10	247	88.8	0.72
11		40	3	6.2	0.5	1.14	301	92.3	0.59
12		40	3	6.2	1.5	1.79	240	90.8	0.76
13		40	7	5.8	0.5	0.81	105	89.5	0.62
14		40	7	5.8	1.5	1.04	113	89.3	0.74
15		40	7	6.2	0.5	1.12	115	88.4	0.60
16		40	7	6.2	1.5	1.83	149	88.6	0.71
17		25	5	6	1	0.83	153	92.9	0.65
18	center	25	5	6	1	0.83	183	93.5	0.74
19		25	5	6	1	0.69	129	91.0	0.66

[a)] DEV-007-01: Expired acetate load/wash buffer lot used in screening run 6. Run retained.
Factors: A, protein load. B, load/Wash conductivity. C, elution buffer pH. D, elution stop collect (AU, OD280, descending).
Not tested in this design: LMW by nrCE-SDS.

Appendix B. Response-surface design matrix and responses

Table B.1 gives the settings and results of every response-surface run. Table B.2 gives residual DNA and leached Protein A in the load and the pool of the same runs.

Table B.1: Cation exchange response-surface design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW	
		A	B	C	D	HMW	HCP	Yield	SEC	nrCE-SDS
		g/L resin	mS/cm	pH	AU	%	ng/mg	%	%	%
1	factorial	10	3	5.8	0.5	0.52	109	98.0	0.55	0.61
2		10	3	5.8	1.5	0.59	114	95.5	0.75	0.71
3		10	3	6.2	0.5	0.57	151	97.3	0.56	0.60
4		10	3	6.2	1.5	0.82	142	94.2	0.72	0.76
5		10	7	5.8	0.5	0.50	59.3	92.1	0.62	0.47
6		10	7	5.8	1.5	0.62	71.9	94.6	0.67	0.76
7		10	7	6.2	0.5	0.69	60.1	93.2	0.59	0.58
8		10	7	6.2	1.5	0.81	73.0	92.5	0.71	0.54
9		40	3	5.8	0.5	0.77	237	87.0	0.65	0.53
10		40	3	5.8	1.5	0.97	254	87.8	0.70	0.67
11		40	3	6.2	0.5	1.15	301	90.0	0.55	0.67
12		40	3	6.2	1.5	1.76	270	89.0	0.69	0.64
13		40	7	5.8	0.5	0.83	174	90.2	0.60	0.56
14[a)]		40	7	5.8	1.5	1.12	139	90.3	0.70	0.73
15		40	7	6.2	0.5	1.08	118	89.1	0.60	0.63
16		40	7	6.2	1.5	1.90	119	84.4	0.75	0.64
17	axial	10	5	6	1	0.52	73.7	93.5	0.62	0.68
18		40	5	6	1	1.11	186	87.9	0.74	0.67
19		25	3	6	1	0.67	225	92.0	0.66	0.60
20		25	7	6	1	0.79	92.9	90.9	0.75	0.54
21		25	5	5.8	1	0.65	149	92.1	0.68	0.63
22		25	5	6.2	1	0.88	148	91.4	0.62	0.59
23		25	5	6	0.5	0.64	162	93.5	0.61	0.60
24		25	5	6	1.5	0.97	150	92.2	0.68	0.67
25	center	25	5	6	1	0.80	146	88.1	0.71	0.70
26		25	5	6	1	0.71	110	95.2	0.61	0.62
27		25	5	6	1	0.78	128	92.1	0.65	0.65
28		25	5	6	1	0.80	152	91.7	0.61	0.55

[a)] DEV-007-02: Column inlet temperature excursion during RSM run 14. Run retained.
Factors: A, protein load. B, load/Wash conductivity. C, elution buffer pH. D, elution stop collect (AU, OD280, descending).

Table B.2: Residual DNA and leached Protein A in the response-surface runs.

Run	DNA load (pg/mg)	DNA pool (pg/mg)	DNA clearance (log10)	Leached Protein A load (ppm)	Leached Protein A pool (ppm)	Leached Protein A clearance factor
1	745	13.6	1.74	3.27	0.27	12.2
2	715	24.4	1.47	3.85	0.30	12.7
3	730	28.8	1.40	3.38	0.34	9.89
4	677	27.8	1.39	3.15	< LOQ	> 12.6
5	722	24.4	1.47	3.67	0.30	12.3
6	664	20.3	1.51	3.70	< LOQ	> 14.8
7	780	29.5	1.42	3.59	0.37	9.72
8	831	28.2	1.47	3.28	0.47	6.96
9	820	17.1	1.68	3.87	0.46	8.44
10	792	20.7	1.58	3.74	< LOQ	> 14.9
11	777	13.9	1.75	3.59	0.44	8.20
12	657	20.0	1.52	3.52	0.49	7.16
13	626	20.0	1.50	3.79	< LOQ	> 15.2
14	753	43.6	1.24	3.54	0.38	9.42
15	558	28.9	1.29	3.59	< LOQ	> 14.3
16	809	14.8	1.74	3.51	0.35	10.1
17	809	21.2	1.58	3.48	0.31	11.4
18	781	29.9	1.42	3.49	0.65	5.36
19	697	17.0	1.61	3.95	0.27	14.7
20	746	13.5	1.74	3.73	0.66	5.64
21	591	25.9	1.36	3.52	0.48	7.29
22	898	12.7	1.85	3.58	< LOQ	> 14.3
23	716	23.6	1.48	3.92	0.44	8.92
24	581	14.4	1.60	3.97	0.35	11.4
25	669	30.2	1.35	3.39	0.29	11.5
26	795	16.2	1.69	3.45	0.29	12.1
27	720	19.3	1.57	3.58	0.55	6.48
28	737	21.3	1.54	3.52	0.53	6.69

LOQ of the qPCR method 1.00 pg/mg and of the leached Protein A ELISA 0.25 ppm. Below the LOQ the clearance factor is the load divided by the LOQ, a lower bound.

Appendix C. Full effect and coefficient tables

Table C.1 gives every effect of the screening models, and Table C.2 the coefficients of the selected step yield model. The coefficients of the aggregate and host cell protein models are in Table 5.6. Fragments have no adequate model, and no coefficient table is given for them.

Table C.1: Effects of the screening models, in the unit of the response.

Response	Term	Effect	SE of coefficient	t	p
Aggregates, HMW (%)	A	0.60	0.02	12.47	< 0.001
	C	0.34	0.02	7.09	< 0.001
	D	0.32	0.02	6.65	< 0.001
	AC	0.20	0.02	4.08	0.004
	AD	0.15	0.02	3.22	0.012
	CD	0.13	0.02	2.74	0.026
	B	−0.02	0.02	−0.38	0.710
	BC	0.02	0.02	0.32	0.760
	BD	−0.01	0.02	−0.27	0.791
	AB	0.01	0.02	0.25	0.808
Host cell protein (ng/mg)	B	−127	8.85	−7.15	< 0.001
	A	116	8.85	6.58	< 0.001
	AB	−56.7	8.85	−3.21	0.012
	BD	39.6	8.85	2.24	0.056
	D	−25.8	8.85	−1.46	0.182
	AD	−24.6	8.85	−1.39	0.202
	CD	24.4	8.85	1.38	0.205
	BC	18.4	8.85	1.04	0.330
	AC	−12.5	8.85	−0.71	0.500
	C	−9.45	8.85	−0.53	0.608
Step yield (points)	A	−5.18	0.29	−9.04	< 0.001
	BC	−1.21	0.29	−2.12	0.067
	B	−1.11	0.29	−1.93	0.090
	AD	−0.91	0.29	−1.59	0.150
	D	0.68	0.29	1.19	0.268
	C	0.67	0.29	1.16	0.279
	BD	−0.57	0.29	−0.99	0.352
	CD	−0.47	0.29	−0.82	0.434
	AC	0.40	0.29	0.70	0.504
	AB	0.00	0.29	0.01	0.995

Effect, twice the coded coefficient, the change across the range of the factor. Letters as in the coded factor table.

Table C.2: Coefficients of the selected step yield model (raw scale, percentage points per coded unit).

Term	Coefficient	SE	t	p
Intercept	91.64	0.33	278.31	< 0.001
A	−3.07	0.41	−7.48	< 0.001
B	−0.76	0.41	−1.84	0.078
AB	0.80	0.44	1.84	0.078

Appendix D. Analytical methods summary

Table D.1 summarises the methods behind the responses of this report.

Table D.1: Analytical methods for the responses of the cation exchange studies.

Analyte	Technique	Method validation	Reportable	Precision	LOQ
Aggregate (HMW)	SEC	AMV-3011	% HMW	SD 0.04 points	not recorded
Fragments (LMW)	nrCE-SDS (SOP-3015)	AMV-3020	% clipped species	SD 0.06 points	0.1 %
Host cell protein	ELISA	AMV-3012	ng per mg antibody	9.5 % RSD	1 ng/mg
Residual DNA	qPCR	AMV-3014	pg per mg antibody	15 % RSD	1 pg/mg
Leached Protein A	ELISA	AMV-3016	ng per mg antibody (ppm)	6.5 % RSD	0.25 ppm
Step yield (antibody mass)	Absorbance at 280 nm	AMV-3019	g/L	1.2 % RSD	0.5 g/L

Titles of the method validations. AMV-3011, Size-Variants (SEC-HPLC). AMV-3012, Host-Cell Protein ELISA. AMV-3014, Residual DNA (qPCR). AMV-3016, Leached Protein A by ELISA (ppm). AMV-3019, Protein Concentration by A280 (UV). AMV-3020, Purity (non-reduced CE-SDS). Precision as an SD in the unit of the result or as a relative SD.

References

Box, George E. P., and David R. Cox. 1964. “An Analysis of Transformations.” *Journal of the Royal Statistical Society, Series B (Methodological)* 26 (2): 211–52.

Burnham, Kenneth P., and David R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. New York: Springer.

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.

Hurvich, Clifford M., and Chih-Ling Tsai. 1989. “Regression and Time Series Model Selection in Small Samples.” *Biometrika* 76 (2): 297–307.

International Council for Harmonisation. 2008. “ICH Q10: Pharmaceutical Quality System.” ICH.

———. 2009. “ICH Q8(R2): Pharmaceutical Development.” ICH.

———. 2012. “ICH Q11: Development and Manufacture of Drug Substances.” ICH.

———. 2023a. “ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin.” ICH.

———. 2023b. “ICH Q9(R1): Quality Risk Management.” ICH.

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. 9th ed. Hoboken, NJ: Wiley.

U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.